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METHODS AND SYSTEMS FOR NONINVASIVE AND LOCALIZED BRAIN LIQUID BIOPSY USING FOCUSED ULTRASOUND

Abstract

Provided herein is a method for diagnosing brain cancer. The method can include applying a focused ultrasound to a brain or brain region of a patient, obtaining a biological sample from the patient, measuring one or more diagnostic fragmentomic features of cfDNA in the biological sample, comparing the one or more diagnostic fragmentomic features to one or more thresholds, and diagnosing the patient with brain cancer when the one or more diagnostic fragmentomic features exceed the one or more thresholds. The focused ultrasound can be operable to disrupt a blood brain barrier of the patient and release cfDNA across the blood brain barrier of the patient. The biological sample can include the cfDNA released across the blood brain barrier.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation-in-part application of U.S. application Ser. No. 17/819,815, filed Aug. 15, 2022, which is a continuation-in-part application of U.S. application Ser. No. 16/393,322 filed 24 Apr. 2019, which claims priority from U.S. Provisional Application Ser. No. 62/662,013 filed on 24 Apr. 2018, all of which are incorporated herein by reference in their entirety.

MATERIAL INCORPORATED-BY-REFERENCE

[0003] The instant application contains a Sequence Listing which has been submitted electronically in Extensible Markup Language (.xml) format and is hereby incorporated by reference in its entirety. Said XML copy, created on Aug. 15, 2022, is named 047563_735896_Sequence_Listing.xml and is 6,395 bytes in size.

FIELD OF THE INVENTION

[0004] The present disclosure generally relates to methods and systems for non-invasive brain biopsies.

SUMMARY OF THE INVENTION

[0005] Among the various aspects of the present disclosure is the provision of a non-invasive and localized brain liquid biopsy using a focused ultrasound system. Briefly, therefore, the present disclosure is directed to improved methods to identify brain tumor or lesion characteristics without the need for a surgical brain biopsy.

[0006] One aspect of the present disclosure provides for methods for non-invasively obtaining a biopsy of a brain tumor or lesion using high-resolution MRI guided focused ultrasound and blood based genetic testing or biomarker detection.

[0007] Another aspect of the present disclosure provides for methods for biopsying or diagnosing a brain tumor or lesion, non-invasively, using MRI guided ultrasound and genetic testing or biomarker detection.

[0008] Another aspect of the present disclosure provides for methods for disrupting brain tumor cells to identify genetic information or using sequencing to identify or diagnose a tumor or lesion.

[0009] Another aspect of the present disclosure provides for methods for using focused ultrasound to release genetic information across the blood brain barrier (BBB) and analyzing the genetic information.

[0010] Another aspect of the present disclosure provides for a method of non-invasively releasing biomarkers from a brain or a brain region across a blood brain barrier (BBB) of a subject.

[0011] In some embodiments, the method comprises applying a focused ultrasound (FUS) to the brain or the brain region, wherein the FUS is applied for a period of time and at an acoustic pressure sufficient to disrupt the BBB and release a detectable quantity of a biomarker across the BBB; obtaining a biological sample from the subject after applying the FUS to a subject brain or a subject brain region; or detecting a biomarker in the biological sample.

[0012] In some embodiments, the biological sample is a biological fluid selected from the group consisting of: blood, cerebral spinal fluid (CSF), interstitial fluid (ISF), serum, and plasma.

[0013] In some embodiments, the period of time sufficient to disrupt the blood brain barrier (BBB) and release a detectable quantity of a biomarker across the BBB is between about 1 minute and

about 4 minutes.

[0014] In some embodiments, the acoustic pressure sufficient to disrupt the blood brain barrier (BBB) and release a detectable quantity of a biomarker across the BBB is between about 0.1 MPa and about 10 MPa.

[0015] In some embodiments, the biomarker comprises genetic material.

[0016] In some embodiments, the genetic material is selected from proteins, extracellular vesicles, cell-free RNA, cell-free DNA, mRNA, circulating tumor DNA (ctDNA), or plasma DNA concentration.

[0017] In some embodiments, the biomarker is selected from D-2-hydroxyglutarate (D2HG) or IDH1((R132H) mutation.

[0018] In some embodiments, the method comprises: scanning a subject head using a magnetic resonance imaging (MRI) scanner and stereotactically coregistering the subject head and identifying a region to be targeted with the FUS.

[0019] In some embodiments, the method comprises assessing the effectiveness of the BBB disruption or release of biomarkers comprising measuring MRI contrast before and after FUS, wherein an increase in MRI contrast after FUS compared to the MRI contrast before FUS indicates successful release of biomarker from the brain or brain region.

[0020] In some embodiments, detecting the biomarker in a biological sample comprises genetic testing or sequencing.

[0021] In some embodiments, the method comprises extracting cell-free or exosomic DNA or RNA from the biological sample.

[0022] In some embodiments, the method comprises detecting somatic mutations in the DNA or RNA using a targeted ultra-deep sequencing technology selected from the group consisting of: ddPCR, AmpliSeq, and HaloPlex sequencing.

[0023] In some embodiments, the method comprises comparing a level of a biomarker in the biological sample after administering the FUS to a biological sample obtained from the subject before FUS or of a matched control sample or standard sample.

[0024] In some embodiments, the brain or brain region comprises a tumor, lesion, or suspected tumor.

[0025] In some embodiments, the subject has or is suspected of having a central nervous system cancer or tumor; a brain tumor, a brain lesion, a neurological disease, disorder, or condition, or a neurodegenerative disease disorder, or condition.

[0026] In some embodiments, the FUS is applied for a period of time and at a pressure sufficient to rupture cells at the region to release biomarkers.

[0027] In some embodiments, the method comprises administering microbubbles to a subject in an amount sufficient to disrupt the BBB upon application of FUS.

[0028] In some embodiments, the method comprises providing an acoustic sensor; or detecting an acoustic signal, wherein the sensor is capable of measuring or monitoring cavitation acoustic emissions.

[0029] Another aspect of the present disclosure provides for a system suitable for use in delivering focused ultrasound (FUS) to a brain or a region of a brain of a subject. In some embodiments, the system comprises a device comprising a plurality of modular pieces configured to deliver ultrasound to the brain or a brain region. In some embodiments, the system comprises an ultrasound transducer configured to emit a FUS beam. In some embodiments, the system comprises an acoustic lens configured to control the FUS beam direction.

[0030] In some embodiments, the system is configured for incorporation into a magnetic resonance imaging (MRI) scanner, wherein the MRI scanner is configured to provide an MRI image for guiding or monitoring of the FUS.

[0031] In some embodiments, the system comprises a helmet configured to fit over a head of a subject.

[0032] In some embodiments, the helmet is demountably coupled to the ultrasound transducer; the ultrasound transducer is demountably coupled to the acoustic lens; the ultrasound transducer is a flat ultrasound transducer; the acoustic lens is a 3D printed acoustic lens; or the acoustic lens is a convex lens or concave lens.

[0033] In some embodiments, the system is suitable for use in parallel with a plurality of the systems.

[0034] In some embodiments, the system comprises a sensor capable of measuring or monitoring cavitation acoustic emissions.

[0035] In some embodiments, the ultrasound transducer is configured to emit a low acoustic pressure of less than about 10 MPa.

[0036] In some embodiments, the system comprises one to eight transducers.

[0037] Further provided herein is a method for diagnosing brain cancer. The method can include applying a focused ultrasound to a brain or brain region of a patient, obtaining a biological sample from the patient, measuring one or more diagnostic fragmentomic features of cfDNA in the biological sample, comparing the one or more diagnostic fragmentomic features to one or more thresholds, and diagnosing the patient with brain cancer when the one or more diagnostic fragmentomic features exceed the one or more thresholds. The focused ultrasound can be operable to disrupt a blood brain barrier of the patient and release cfDNA across the blood brain barrier of the patient. The biological sample can include the cfDNA released across the blood brain barrier.

[0038] In some aspects, the one or more diagnostic fragmentomic features can include one or more of a ratio of short fragments to long fragments, a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes.

[0039] In some aspects, the method can further include obtaining a baseline biological sample prior to applying the focused ultrasound. In some aspects, the one or more baseline fragmentomic features can include one or more of a ratio of short fragments to long fragments, a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes.

[0040] In some aspects, the one or more thresholds can include a percentage increase or decrease between the one or more baseline fragmentomic features and the one or more diagnostic fragmentomic features. In some aspects, the one or more thresholds can be empirically determined thresholds from a population of healthy patients and/or a population of patients with brain cancer.

[0041] In some aspects, the biological sample can include a blood sample. In some aspects, the focused ultrasound can be applied for a period of time sufficient to disrupt the blood brain barrier. In some aspects, the period of time can be about 1 minute to about 4 minutes. In some aspects, the acoustic pressure can be about 0.1 MPa to about 10 MPa.

[0042] In some aspects, the method can further include assessing an effectiveness of disruption of the blood brain barrier or release of cfDNA by measuring MRI contrast before and after applying the focused ultrasound, wherein an increase in the MRI contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

[0043] In some aspects, the method can further include assessing an effectiveness of disruption of the blood brain barrier or release of cfDNA by measuring CT contrast before and after applying the focused ultrasound, wherein an increase in the CT contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

[0044] In some aspects, the method can further include assessing an effectiveness of disruption of the blood brain barrier or release of cfDNA by measuring ultrasound contrast before and after applying the focused ultrasound, wherein an increase in the ultrasound contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

[0045] In some aspects, the brain or brain region can include a suspected tumor. In some aspects, the method can further include administering microbubbles to the patient in an amount sufficient to disrupt the blood brain barrier when applying the focused ultrasound. In some aspects, the method

can further include detecting, via an acoustic sensor, an acoustic signal. In some aspects, the acoustic sensor can be operable to measure and/or monitor cavitation acoustic emissions. In some aspects, the cfDNA can include ctDNA.

[0046] Other objects and features will be in part apparent and in part pointed out hereinafter.

Description

DESCRIPTION OF THE DRAWINGS

[0047] Those of skill in the art will understand that the drawings, described below, are for illustrative purposes only. The drawings are not intended to limit the scope of the present teachings in any way.

[0048] FIG. 1A-1F. Experimental method. (A) Schematic illustration of the focused ultrasound (FUS) experiment setup for the treatment of U87 tumor-bearing mice. (B) Bioluminescence image of the orthotopic mouse model with the green fluorescence image of the mouse brain shown on the right. (C) Schematic illustration of the MR-guided focused ultrasound (MRgFUS) system for the treatment of GL261 tumor-bearing mice. (D) Representative contrast-enhanced MR images acquired before and after FUS treatment. The enhanced accumulation of the MR contrast agents in the tumor region verified accurate tumor targeting by the FUS. (E) Diagram of FUS pulses. (F) Illustration of the experimental timeline.

[0049] FIG. 2A-2C. Comparison of the circulating eGFP mRNA expression in the control and treated U87 mice. (A) Amplification curves [$\log_{10}(\text{sub.2}(\Delta R_n))$] of circulating eGFP mRNA in the control (n=3) and treated mice (n=6) for two primer pairs, eGFP A and eGFP B. ΔR_n is the fluorescence intensity of eGFP mRNA minus the baseline. Comparison of the relative expression levels ($2^{\text{sup.}-\Delta C_{\text{sub.T}}}$) of (B) eGFP A and (C) eGFP B in the control and treated mice. *: $p < 0.05$

[0050] FIG. 3A-3B. Comparison of the circulating eGFP mRNA expression in the control and treated GL261 mice. Comparison of the expression levels ($2^{\text{sup.}-\Delta C_{\text{sub.T}}}$) of (A) eGFP A and (B) eGFP B in the control group and three treatment groups with different acoustic pressures: 1.48 MPa, 2.10 MPa, and 3.34 MPa. All the measured data points as well as their mean and standard deviation are shown for each group. The circulating mRNA levels of eGFP A and eGFP B were significantly higher in the FUS-treated groups compared with the untreated control group (eGFP A: $p = 0.0045$; eGFP B: $p = 0.0045$). The expression levels of mice in the 1.48 MPa group was significantly higher than those of the other two groups for eGFP A and eGFP B (eGFP A: $p = 0.012$; eGFP B: $p = 0.012$).

[0051] FIG. 4A-4D. Histological assessment of brain tissue from the control and treated GL261 mice. H&E staining of the ex vivo tumor slices obtained from the control mice (A) and mice treated with FUS at (B) 1.48 MPa, (C) 2.10 MPa, and (D) 3.34 MPa, respectively. Hemorrhages were observed in all the FUS treated mice.

[0052] FIG. 5A-5C. Experimental setup. (A) Illustration and (B) picture of the MR-guided FUS system used for the FUS-LBx treatment. A clinical MR-guided FUS system that integrated a clinical MRI scanner with a FUS phased array was configured for small animal study by coupling with a small animal adaptor (see Methods for more details). (C) MR image of a fiber-optic hydrophone used for measuring the acoustic pressures. The hydrophone was placed vertically in a water container that was on top of the small animal adapter. The FUS transducer's focus (red dot) was aligned at the tip of the optical fiber for pressure measurements.

[0053] FIG. 6A-6B. Effect of peak negative pressure on FUS-LBx yield. Comparison of the circulating eGFP mRNA level in the control mice receiving no treatment and FUS-treated mice at 0.59, 1.29, and 1.58 MPa. eGFP mRNA levels were quantified using quantitative polymerase chain reaction (qPCR) with (A) primer A and (B) primer B.

[0054] FIG. 7A-7F. Effect of peak negative pressure on brain hemorrhage. (A-D) Representative

images of H&E-stained sections from mice in the control group, 0.59 MPa, 1.29 MPa, and 1.58 MPa groups, respectively. Boxes in the images on the top indicate the location where the higher magnification images (bottom) were acquired. (E) Illustration of microhemorrhage quantification using the positive pixel detection algorithm (see Methods). Yellow lines highlight the detected microhemorrhage. (C) Microhemorrhage density is significantly higher in mice treated with higher pressures (1.29 MPa and 1.58 MPa) compared to the low pressure (0.59 MPa) and control mice.

[0055] FIG. 8A-8E. Relationships between intratumoral and peritumoral MR image contrast enhancements and microhemorrhage density and eGFP mRNA level. (A) Overview of MRI analysis workflow. Tumor MR contrast enhancement was calculated for intratumoral area and peritumoral area (defined as the brain region within 2 mm vicinity of the tumor). (B) Correlation analyses between intratumoral and peritumoral enhancements with microhemorrhage density and plasma eGFP mRNA level in FUS-treated mice (n=15) using primer A.

[0056] FIG. 9 is a photo of the transducer.

[0057] FIG. 10 is a photo of a mouse treated using the device.

[0058] FIG. 11 is a photo showing the successful drug delivery to the mouse brain, indicated by the leakage of the blue color.

[0059] FIG. 12 is an image of spatially precise drug delivery to the hippocampus, indicated by the enhanced fluorescence observed on the right side of the of the mouse hippocampus.

[0060] FIG. 13 is a series of images of an MRgFUS system design for a pig.

[0061] FIG. 14. Planning and targeting of image-guided focused ultrasound liquid biopsy (FUS-LBx).

[0062] FIG. 15. Monitoring and feedback control using an US sensor to monitor microbubble response to US.

[0063] FIG. 16. Outcome assessment with MRI.

[0064] FIG. 17. FUS-LBx in a large animal model (pig). GFAP: Glial fibrillary acidic protein; MBP: Myelin basic protein; S100B: S100 calcium-binding protein B.

[0065] FIG. 18A-18F show MRI-guided sonobiopsy changed composition of circulating phosphorylated tau species in 2-month-old PS19 mice (Experiment 1). (A) Diagram of the FUS system in the small animal MRI scanner. The mouse head was fixed in the MRI coil and the FUS transducer was coupled to the skull with ultrasound gel and a water balloon filled with degassed water. (B) Schematic of the FUS trajectory targeting the right cerebral hemisphere. (C) T₂-weighted MRI scans were acquired before FUS. (D) Post-FUS T₂-weighted MRI scans confirmed FUS-induced BBB disruption as a signal enhancement. There was no significant group difference (p=0.71) in the volume of FUS-mediated BBB opening between PS19 (30.9±17.9 mm³) and wild-type mice (35.1±22.9 mm³). (E) In PS19 mice, the normalized pTau-181 (pTau-181/mTau ratio) was significantly greater (p=0.0058) in the sonobiopsy group (n=6; 0.57±0.19) compared with the normalized pTau-181 in the control group (n=6; 0.36±0.09). (F) In PS19 mice, the normalized pTau-231 (pTau-231/mTau ratio) was significantly greater (p=0.048) in the sonobiopsy group (n=6; 0.17±0.06) compared with the normalized pTau-181 in the control group (n=6; 0.13±0.03).

[0066] FIG. 19A-19C show absolute levels of pTau-181, pTau-231, and mTau. (A) The plasma level of pTau-181 in the PS19 mice (control: 1523.99±1486.82 pg/mL; sonobiopsy: 1643.18±972.79 pg/mL) was higher than the wild-type mice (control: 0.47±0.22 pg/mL; sonobiopsy: 0.52±0.17 pg/mL). There was no significant difference in the absolute plasma level of pTau-181 in either the PS19 (p=0.56) or the wild-type mice (p=0.64). (B) The absolute level of pTau-231 in the plasma of the PS19 mice (control: 628.05±873.03 pg/mL; sonobiopsy: 806.95±1002.73 pg/mL) was higher than the wild-type mice (control: 81.91±41.83 pg/mL; sonobiopsy: 118.03±57.13 pg/mL). There was no significant difference in the absolute plasma level of pTau-231 in either the PS19 (p=0.64) or the wild-type mice (p=0.89). (C) The absolute level of mTau was greater in the PS19 mice (control: 4403.10±5393.20 pg/mL; sonobiopsy:

4669.61±4831.65 pg/mL) compared with the wild-type mice (control: 306.79±155.02 pg/mL; sonobiopsy: 458.35±246.66 pg/mL). There was no significant difference in the absolute plasma level of mTau in either the PS19 (p=0.54) or the wild-type mice (p=0.90).

[0067] FIG. 20A-20B show normalized pTau-181 and pTau-231 levels for wild-type mice. (A) The normalized level of pTau-181 in the sonobiopsy group ($1.4 \times 10^{-3} \pm 0.9 \times 10^{-3}$) was not significantly different (p=0.71) from that in the control group ($1.9 \times 10^{-3} \pm 1.8 \times 10^{-3}$). (B) The normalized level of pTau-231 in the sonobiopsy group (0.27 ± 0.05) was not significantly different (p=0.55) from that in the control group (0.27 ± 0.06).

[0068] FIG. 21A-21H show sonobiopsy enhanced levels of plasma NfL in 6-month-old PS19 mice (Experiment 2). (A) Schematic of the FUS trajectory targeting the olfactory cerebral cortex (piriform cortex and amygdala). (B) T_{sub.1}-weighted MRI scans were acquired before FUS. (C) Post-FUS T_{sub.1}-weighted MRI scans confirmed FUS-induced BBB disruption as a signal enhancement. (D) Schematic of the FUS trajectory targeting the hippocampus. (E) T_{sub.1}-weighted MRI scans were acquired before FUS. (F) FUS T_{sub.1}-weighted MRI scans confirmed FUS-induced BBB disruption as a signal enhancement. There was no significant difference (p=0.17) in the volume of FUS-mediated BBB disruption between mice treated at the olfactory cerebral cortex (21.7 ± 5.8 mm^{sup.3}) or the hippocampus (15.4 ± 5.8 mm^{sup.3}). (G) Sonobiopsy resulted in a 2.3-fold (± 0.78) increase in plasma NfL levels (n=8; $p = 2.2 \times 10^{-4}$). (H) The normalized plasma NfL (post-FUS/pre-FUS) was significantly greater (p=0.011) in the sonobiopsy group (n=9; 3.34 ± 2.15) than the control group (n=5; 1.28 ± 0.22).

[0069] FIG. 22A-22D show a safety assessment of sonobiopsy. (A) Representative H&E staining for experiment 1 with 2-month-old mice. (B) There was a significant increase (p=0.04) in microhemorrhage density (red arrow) in the treated hemisphere ($1.38 \pm 0.92\%$) compared with the contralateral hemisphere ($0.48 \pm 0.44\%$). (C) Representative H&E staining for experiment 2 with 6-month-old mice. (D) There was no significant difference (p=0.20) in the microhemorrhage density in the treated hemisphere ($0.52 \pm 0.32\%$) compared with the contralateral hemisphere ($0.42 \pm 0.29\%$).

[0070] FIG. 23A-23B show a safety assessment of sonobiopsy via histological examination. (A) In experiment 1, there was a non-significant (p=0.07) sub-millimeter increase in average microhemorrhage area in the treated hemisphere (0.051 ± 0.038 mm^{sup.2}) compared with that in the contralateral hemisphere (0.017 ± 0.018 mm^{sup.2}). (B) In experiment 2, there was no significant difference (p=0.39) in the average microhemorrhage area in the treated hemisphere (0.014 ± 0.006 mm^{sup.2}) compared with the contralateral hemisphere (0.013 ± 0.017 mm^{sup.2}).

[0071] FIG. 24A-24B show a safety assessment of sonobiopsy via passive cavitation detection. (A) In experiment 1, the inertial cavitation dose was minimal with no significant difference (p=0.67) between the FUS-treated groups (PS19: -0.14 ± 0.68 dB; wild-type: 0.007 ± 0.623 dB), suggesting a low risk for tissue damage. (B) In experiment 2, there was minimal inertial cavitation dose during sonication of the 6-month-old PS19 mice at the piriform cortex (-0.045 ± 0.056 dB) and the hippocampus (0.008 ± 0.089 dB).

[0072] FIG. 25 is a flowchart of a method for diagnosing brain cancer in one example.

DETAILED DESCRIPTION OF THE INVENTION

[0073] The present disclosure is based, at least in part, on the discovery that a “liquid brain biopsy” can provide similar levels of information as a standard brain biopsy. As described herein, it was demonstrated that the combination of FUS and microbubbles allows for the detection of biomarkers of various diseases or tumor-specific mRNA in the bloodstream that is otherwise undetectable. These presently disclosed findings established that FUS-mediated BBB disruption could enhance brain-to-blood trafficking. FUS may offer an enabling technique for a noninvasive and regionally-specific brain tumor liquid biopsy that can be utilized in personalized brain cancer or neurodegenerative disease patient care.

[0074] In typical scenarios when a brain lesion is identified, or when a patient has a known brain tumor that has been previously treated, there is a clinical indication to perform a surgical

stereotactic biopsy. This entails making a hole in the skin and skull and placing a needle probe into the lesion and aspirating a piece of tissue for histological and genetic testing to achieve a diagnosis. Also of note, there are a number of other lesions in the brain that can potentially require biopsy—lesions of unknown etiology, radiation necrosis, or known tumors, which require further clarification if genetic profile has changed, etc. This procedure can carry the surgical risks of bleeding, infection, non-diagnosis, or the requirements of general anesthesia.

[0075] This invention provides similar levels of information without the requirement for surgery. The general concept involves the following process. A lesion in the brain is identified that requires pathologic diagnosis. The lesion is imaged with MRI using high-resolution anatomic imaging (i.e., T1 and T2 sequences). The imaging is stereotactically coregistered to the patient's head using standard frameless navigation systems. In some examples, stereotactically coregistering to the patient's head can use an MRI machine, CT scanner, or ultrasound imaging system. Targets for biopsy are then identified with the navigation system. Once done, a focused ultrasound source is then used to target the region. The patient is concurrently given ultrasound fluoridated microbubbles (microbubbles can be optional). The focused ultrasound or combination of the ultrasound and microbubbles can induce a disruption of the blood brain barrier in the prospectively chosen target site. An alternative to the fluoridated microbubbles would be to simply have higher power focused ultrasound that creates a local tissue injury and disruption of the blood brain barrier. This US perturbation can also lead to the rupture of the cells and the release of genetic material into the blood stream. Biological samples (e.g., blood) and, optionally, matched normal samples can then be taken from the patient and genetically analyzed in order to identify biomarkers, such as circulating somatic variants. These somatic variants can then be evaluated to determine the nature of the lesion in the brain.

[0076] The technology/process can involve the novel combination of focused ultrasound and advanced genetic screening. Specifically, the ultrasound technique can be a simple, compact, and relatively inexpensive focused ultrasound system that is comprised of a single-element FUS transducer. The transducer can be coupled to a high-precision, fast-speed positioning system to steer transducer focus mechanically. It is also an option to use a focused ultrasound system comprising a phased array transducer with multiple elements for steering the transducer focal point electronically.

[0077] These techniques can be used in conjunction with either stereotactic navigation in which the patient's brain imaging and lesion localization is rigorously coregistered in physical geometric space. This can allow for precise perturbation of the intended target. Once performed, blood or other fluid can be withdrawn from the patient and cell-free or exosomic DNA and/or RNA or other biomarkers can be extracted. In an example, the DNA and/or RNA can be evaluated for the presence of somatic mutations or somatic variants using a targeted ultra-deep sequencing technology, examples of which include, but are not limited to, PCR, qPCR, ddPCR, AmpliSeq, or HaloPlex sequencing. In some examples, the DNA and/or RNA can be evaluated for the presence of somatic mutations or somatic variants using non-sequencing based PCR techniques, such as ddPCR. The set of variants to be queried can contain patient-specific variants (identified using whole-genome or whole-exome sequencing of a surgically-resected tumor and matched normal tissue from the same patient), a panel of variants that are commonly present in brain tumors or a combination of panel variants and patient-specific variants. Optionally, genomic DNA from matched normal samples from the same patient can be sequenced and used to distinguish somatic variants from germline variants.

[0078] In addition, these techniques can be used to release and identify other biomarkers present in the brain, for example, biomarkers for neurodegenerative diseases.

[0079] In some examples, the systems and methods described herein may be used in combination with diagnostic imaging to improve the yield or recovery of biomarker. For example, a specific location in the brain for applying the FUS may be determined with the use of diagnostic imaging.

Non-limiting examples of diagnostic imaging include MRI, PET, SPECT, and ultrasound.

[0080] It is also important to note that while blood is the easiest and most accessible source of fluid to perform further genetic analysis or biomarker detection, one could also perform this same analysis on cerebrospinal fluid (CSF) or other biological samples (e.g., blood, plasma, serum, interstitial fluid (ISF)).

Blood Brain Barrier (BBB) Opening with Focused Ultrasound (FUS)

[0081] The blood-brain barrier (BBB) blocks large molecules (>400 Da) from entering the central nervous system (CNS). Focused Ultrasound (FUS) is the only available technique to non-invasively, locally, and transiently open the BBB. Previous preclinical research using small animal models has demonstrated this technique as a promising way to treat various brain disorders. FUS in combination with microbubbles can also be used. This technique has also been tested on non-human primates with success. Clinical Studies with FUS are actively investigated for treatment of brain tumor and Alzheimer's disease. Besides its clinical translation potential, this technique has a broad application in preclinical research. This technique can be used to deliver various agents to the brain, such as chemo drugs, proteins, peptides, nanoparticles, gene vectors, and stem cells. However, currently there is a lack of devices that are affordable, high throughput, and dedicated for small animal research. Here is disclosed a FUS-induced BBB opening device that meets these needs.

[0082] The BBB, as described herein, can also comprise a blood-tumor barrier. The blood-tumor barrier is less leaky than vessels outside the central nervous system, but is leakier than a healthy BBB.

[0083] As described herein, the BBB can be opened for minutes, hours, or days, depending on the US energy administered to the subject. This can allow for serial testing of biomarkers. It is preferred to have as short of an open window as possible (e.g., on the order of minutes) because of the possibility of infection or because the detectable biomarkers can have a short half-life, are unstable, or are quickly cleared from the bloodstream. Because the half-life for many biomarkers is known, the time point for collection is known as well (e.g., the time at which a sample is taken from a subject can correlate to a time where the biomarker has not yet been cleared from the subject's system).

Focused Ultrasound (FUS) System

[0084] Disclosed herein is a focused ultrasound (FUS) system for use in disruption of a blood brain barrier (BBB), releasing biomarkers from a brain lesion or tumor. After the subject is administered the ultrasound, the biomarkers can be detected in a biological sample of a subject. Conventional FUS systems being used to open up the BBB are being used with therapeutically-relevant acoustic pressures (e.g., higher peak negative acoustic pressure (PNP) levels than used in typical diagnostic or imaging levels). For example, clinical FUS systems use relatively high acoustic pressure. The FUS systems and methods as described herein can use diagnostically-relevant acoustic pressures to open the blood brain barrier and release biomarkers from a brain lesion or suspected brain lesion.

[0085] As an example, the acoustic pressure for mice in the presently disclosed system can be about 0.6 MPa. As another example, the acoustic pressure for a human in the presently disclosed system can be about 1.5 MPa.

[0086] Furthermore, conventional clinical systems use high energy and require high precision for use in delivering treatment to a specific area, such as a tumor or lesion. The FUS system described herein does not require high energy or precision to disrupt the BBB and release biomarkers from the brain or a brain lesion. For example, a clinical FUS system can comprise over 1,000 elements (a transducer array) that are individually steered. The system described herein only requires a single element (but up to about 8 or up to about 16 can be useful) at diagnostically-relevant energy requirements. This design decreases the cost, simplifies the design, can be adapted for human or animal use, and is designed to be incorporated into any MRI scanner.

[0087] Described herein in also a method of monitoring cavitation (e.g., microbubble behavior)

(see e.g., Example 3). The FUS system can comprise a sensor that measures acoustic signal. In some examples, the FUS system may include more than one sensor, such as an array of sensors. This monitoring can inform if the acoustic pressure should be increased or decreased by measuring the acoustic signal.

[0088] As described herein, the focused ultrasound-enabled brain tumor liquid biopsy (FUS-LBx) offers a technique that can provide noninvasive, spatially resolved, and temporally controlled brain liquid biopsies. FUS has emerged as a technology with the potential to noninvasively exert mechanical and thermal effects on the brain tissue. When coupled with intravenously injected microbubbles, low-intensity pulsed FUS-induced mechanical effects can transiently and non-invasively open the BBB without causing any or substantial vascular or tissue damage. It was demonstrated (see e.g., Example 1) that FUS in combination with microbubbles could also be exploited to release biomarkers from brain tumors in murine glioblastoma models. The approach described here, is different from previously reported FUS-facilitated liquid biopsy techniques developed for in vivo biomarker release from cancers outside the brain. In one of those reports, pulsed high-intensity focused ultrasound (HIFU) with high acoustic pressures (ultrasound frequency=1.5 MHz, peak compressional focal pressure=90 MPa, and peak rarefactional focal pressure=17 MPa) was used to induce histotripsy (i.e., a technique for mechanical tissue fractionation) in a rat model of prostate cancer, and this enhanced the release of cell-free tumor microRNA into the blood circulation. In another study, a chicken embryo tumor model was used to show the feasibility of amplifying the release of extracellular vesicles using HIFU (ultrasound frequency=1.15 MHz and peak to peak pressure was within the range of 10-30 MPa) in combination with phase-change nanodroplets which changed to microbubbles upon HIFU sonication. In a recent study, two protein biomarkers were found to be significantly increased in the plasma of patients undergoing HIFU thermal ablation (ultrasound frequency=1.1 MHz and power of 100-200 W) of uterine fibroids. All these previous studies used HIFU to induce permanent mechanical or thermal disruption of the tumor to enhance the release of tumor biomarkers. The tissue damaging effect limits the application of these techniques as diagnostic tools in a sensitive organ such as the brain. In contrast, the FUS-LBx technique disclosed herein uses low-intensity pulsed FUS, which has the potential advantage of enabling the biomarker release without causing tissue damage. As shown in Example 1, the acoustic pressures used were intentionally selected to be relatively high (1.52-3.53 MPa in mouse model) to increase the chance of success in releasing biomarkers. As expected, hemorrhage was found in the relatively higher intensity FUS-targeted brain area.

[0089] Low-intensity focused ultrasound can be used with low acoustic pressures (e.g., less than about 10 MPa), ultrasound frequency less than about 1.5 MHz, peak compressional focal pressure less than about 90 MPa, or peak rarefactional focal pressure less than about 17 MPa. Low-intensity focused ultrasound can be used with a peak to peak pressure less than about 10 to 30 MPa; an ultrasound frequency less than 1.1 MHz; a power of less than about 100 to 200 W; or peak negative acoustic pressure (PNP) levels of less than about 3.53 MPa.

[0090] Example 2 describes the effects of FUS acoustic pressure on the level of tumor biomarker release and the extent of associated hemorrhage in order to optimize the FUS-LBx technique. Example 2 determined the optimal ultrasonic pressure for FUS-LBx that can sufficiently increase plasma levels of the tumor biomarkers while at the same time minimize the risk of hemorrhage in the brain. It was further explored whether post-sonication changes in tumor MR contrast enhancement can predict successful biomarker release for the future development of image-guided FUS-LBx technique.

[0091] Disclosed herein are systems comprising a FUS-induced BBB opening device that are affordable, high throughput, and can be easily translated for clinical use or dedicated for animal (e.g., pig) or small animal (e.g., mouse) research. The transducer, as described herein can be an inexpensive flat ultrasound transducer coupled to a 3D printed acoustic lens (or acoustic window).

In some examples, the transducer can be a curved ultrasound transducer. The acoustic lens may be a convex lens or a concave lens. The cost of the transducer, as described in the Examples, was about \$80, while the FUS transducer commonly used in preclinical research can cost over \$10,000. Multiple transducers can be used simultaneously to achieve high throughput treatment of multiple animals at the same time.

[0092] A wearable helmet can be used for non-invasive, targeted FUS sonication of the human or animal brain while animals are awake. The helmet can have a modular design, featuring easy removal and installation of the unit for targeting different brain regions.

[0093] The performance of the helmet in inducing BBB opening can be assessed by co-injecting Evan's blue with microbubbles to evaluate BBB permeability using fluorescence imaging of ex vivo brain slices in animals.

[0094] The helmet and ultrasound transducer and wirings are lightweight at about 6.6 g. The constraint design of the helmet can minimize the effect of mouse movement on targeting. The helmet system achieved localized BBB opening at the targeted brain location. The fluorescence intensity of the Evan's blue in the brains of awake mice was higher than that in anesthetized mice, suggesting that FUS-induced BBB opening was affected by anesthesia.

[0095] The helmet design of the FUS device provides an innovative tool to study FUS-induced BBB opening in subjects such as an awake mouse.

[0096] As described herein, the FUS system can use acoustic pressures between about 0.1 MPa and about 10 MPa. For example, the acoustic pressure can be about 0.1 MPa; about 0.2 MPa; about 0.3 MPa; about 0.4 MPa; about 0.5 MPa; about 0.6 MPa; about 0.7 MPa; about 0.8 MPa; about 0.9 MPa; about 1 MPa; about 1.1 MPa; about 1.2 MPa; about 1.3 MPa; about 1.4 MPa; about 1.5 MPa; about 1.6 MPa; about 1.7 MPa; about 1.8 MPa; about 1.9 MPa; about 2 MPa; about 2.1 MPa; about 2.2 MPa; about 2.3 MPa; about 2.4 MPa; about 2.5 MPa; about 2.6 MPa; about 2.7 MPa; about 2.8 MPa; about 2.9 MPa; about 3 MPa; about 3.1 MPa; about 3.2 MPa; about 3.3 MPa; about 3.4 MPa; about 3.5 MPa; about 3.6 MPa; about 3.7 MPa; about 3.8 MPa; about 3.9 MPa; about 4 MPa; about 4.1 MPa; about 4.2 MPa; about 4.3 MPa; about 4.4 MPa; about 4.5 MPa; about 4.6 MPa; about 4.7 MPa; about 4.8 MPa; about 4.9 MPa; about 5 MPa; about 5.1 MPa; about 5.2 MPa; about 5.3 MPa; about 5.4 MPa; about 5.5 MPa; about 5.6 MPa; about 5.7 MPa; about 5.8 MPa; about 5.9 MPa; about 6 MPa; about 6.1 MPa; about 6.2 MPa; about 6.3 MPa; about 6.4 MPa; about 6.5 MPa; about 6.6 MPa; about 6.7 MPa; about 6.8 MPa; about 6.9 MPa; about 7 MPa; about 7.1 MPa; about 7.2 MPa; about 7.3 MPa; about 7.4 MPa; about 7.5 MPa; about 7.6 MPa; about 7.7 MPa; about 7.8 MPa; about 7.9 MPa; about 8 MPa; about 8.1 MPa; about 8.2 MPa; about 8.3 MPa; about 8.4 MPa; about 8.5 MPa; about 8.6 MPa; about 8.7 MPa; about 8.8 MPa; about 8.9 MPa; about 9 MPa; about 9.1 MPa; about 9.2 MPa; about 9.3 MPa; about 9.4 MPa; about 9.5 MPa; about 9.6 MPa; about 9.7 MPa; about 9.8 MPa; about 9.9 MPa; or about 10 MPa. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0097] As described herein, the FUS system can use a duty cycle between about 0.1% and about 10%. For example, the duty cycle can be about 0.1%; about 0.2%; about 0.3%; about 0.4%; about 0.5%; about 0.6%; about 0.7%; about 0.8%; about 0.9%; about 1%; about 1.1%; about 1.2%; about 1.3%; about 1.4%; about 1.5%; about 1.6%; about 1.7%; about 1.8%; about 1.9%; about 2%; about 2.1%; about 2.2%; about 2.3%; about 2.4%; about 2.5%; about 2.6%; about 2.7%; about 2.8%; about 2.9%; about 3%; about 3.1%; about 3.2%; about 3.3%; about 3.4%; about 3.5%; about 3.6%; about 3.7%; about 3.8%; about 3.9%; about 4%; about 4.1%; about 4.2%; about 4.3%; about 4.4%; about 4.5%; about 4.6%; about 4.7%; about 4.8%; about 4.9%; about 5%; about 5.1%; about 5.2%; about 5.3%; about 5.4%; about 5.5%; about 5.6%; about 5.7%; about 5.8%; about 5.9%; about 6%; about 6.1%; about 6.2%; about 6.3%; about 6.4%; about 6.5%; about 6.6%; about 6.7%; about 6.8%; about 6.9%; about 7%; about 7.1%; about 7.2%; about 7.3%; about 7.4%; about 7.5%; about 7.6%; about 7.7%; about 7.8%; about 7.9%; about 8%; about 8.1%; about 8.2%; about 8.3%; about

8.4%; about 8.5%; about 8.6%; about 8.7%; about 8.8%; about 8.9%; about 9%; about 9.1%; about 9.2%; about 9.3%; about 9.4%; about 9.5%; about 9.6%; about 9.7%; about 9.8%; about 9.9%; or about 10%. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0098] As described herein, the FUS system can use a pulse repetition frequency between about 0.1 Hz and about 10 Hz. For example, the pulse repetition frequency can be about 0.1 Hz; about 0.2 Hz; about 0.3 Hz; about 0.4 Hz; about 0.5 Hz; about 0.6 Hz; about 0.7 Hz; about 0.8 Hz; about 0.9 Hz; about 1 Hz; about 1.1 Hz; about 1.2 Hz; about 1.3 Hz; about 1.4 Hz; about 1.5 Hz; about 1.6 Hz; about 1.7 Hz; about 1.8 Hz; about 1.9 Hz; about 2 Hz; about 2.1 Hz; about 2.2 Hz; about 2.3 Hz; about 2.4 Hz; about 2.5 Hz; about 2.6 Hz; about 2.7 Hz; about 2.8 Hz; about 2.9 Hz; about 3 Hz; about 3.1 Hz; about 3.2 Hz; about 3.3 Hz; about 3.4 Hz; about 3.5 Hz; about 3.6 Hz; about 3.7 Hz; about 3.8 Hz; about 3.9 Hz; about 4 Hz; about 4.1 Hz; about 4.2 Hz; about 4.3 Hz; about 4.4 Hz; about 4.5 Hz; about 4.6 Hz; about 4.7 Hz; about 4.8 Hz; about 4.9 Hz; about 5 Hz; about 5.1 Hz; about 5.2 Hz; about 5.3 Hz; about 5.4 Hz; about 5.5 Hz; about 5.6 Hz; about 5.7 Hz; about 5.8 Hz; about 5.9 Hz; about 6 Hz; about 6.1 Hz; about 6.2 Hz; about 6.3 Hz; about 6.4 Hz; about 6.5 Hz; about 6.6 Hz; about 6.7 Hz; about 6.8 Hz; about 6.9 Hz; about 7 Hz; about 7.1 Hz; about 7.2 Hz; about 7.3 Hz; about 7.4 Hz; about 7.5 Hz; about 7.6 Hz; about 7.7 Hz; about 7.8 Hz; about 7.9 Hz; about 8 Hz; about 8.1 Hz; about 8.2 Hz; about 8.3 Hz; about 8.4 Hz; about 8.5 Hz; about 8.6 Hz; about 8.7 Hz; about 8.8 Hz; about 8.9 Hz; about 9 Hz; about 9.1 Hz; about 9.2 Hz; about 9.3 Hz; about 9.4 Hz; about 9.5 Hz; about 9.6 Hz; about 9.7 Hz; about 9.8 Hz; about 9.9 Hz; or about 10 Hz. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0099] As described herein, the FUS system can use an exposure duration between about 1 second and about 20 minutes. For example, the exposure duration can be about 30 s; about 1 min; about 2 min; about 3 min; about 4 min; about 5 min; about 6 min; about 7 min; about 8 min; about 9 min; about 10 min; about 11 min; about 12 min; about 13 min; about 14 min; about 15 min; about 16 min; about 17 min; about 18 min; about 19 min; or about 20 min. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0100] As described herein, a biological sample can be collected at any time prior to or after FUS. For example, the sample collection can correspond with a time less than the half-life of the biomarker to be tested. As another example, biological samples can be serially collected at various time points before or after FUS. As another example, the biological sample can be collected within 10 minutes to about 2 days after FUS exposure or administration. For example, the biological sample can be collected post-FUS at about 1 min, about 2 min, about 3 min; about 4 min; about 5 min; about 6 min; about 7 min; about 8 min; about 9 min; about 10 min; about 11 min; about 12 min; about 13 min; about 14 min; about 15 min; about 16 min; about 17 min; about 18 min; about 19 min; or about 20 min. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0101] As described herein, the system can be used for any subject with a central nervous system and a blood brain barrier. For example, the subject can be an animal subject, including a mammal, such as horses, cows, dogs, cats, sheep, pigs, mice, rats, monkeys, hamsters, guinea pigs, and chickens, and humans. As another example, the subject can be a human subject.

Biological Sample

[0102] The methods and systems designed herein can release biomarkers or genetic material from a brain into a biological sample, such as a biological fluid. The biological fluid or biological sample can be whole blood, blood, plasma, serum, cerebral spinal fluid (CSF), or interstitial fluid (ISF).

Microbubbles

[0103] The methods described herein can comprise the use of microbubbles. Microbubbles are

widely used for the heart and liver. Microbubbles are commercially available (e.g., Definity®). Microbubbles can be used with FUS to provide lower levels of US intensity with similar results to high-intensity US with less potential side effects or damage. Use of microbubbles, generally, is well known; see e.g. Blomley et al. (2001) “Science, medicine, and the future: Microbubble contrast agents: A new era in ultrasound” *BMJ* 322 (7296): 1222-5; Sirsi et al. 2009 *Bubble Sci Eng Technol* 1(1-2): 3-17. Except as otherwise noted herein, therefore, the process of the present disclosure can be carried out in accordance with such processes.

[0104] Microbubbles can be administered by methods currently known in the art. As described herein, the microbubbles are administered according to the clinical dose currently used in treatments (e.g., for Definity® microbubbles, 1.3 mL from a diluted vial is diluted with 8.7 mL saline).

[0105] Microbubbles are small, gas-filled bubbles, which can be between 0.5 μm and 10 μm in diameter. Microbubbles are widely used as contrast agents in medical imaging and as carriers for targeted drug delivery. The core of the microbubble is a gas, which is surrounded by a shell that can be composed of polymers, lipids, lipopolymers, proteins, surfactants, or a combination of these.

[0106] Microbubbles can be injected intravenously, a process that researchers have shown is safe compared to the use of conventional contrast agents in techniques such as magnetic resonance imaging and radiography.

[0107] Microbubbles resonate in an ultrasound beam, contracting and expanding as pressure changes occur in the ultrasound wave. Microbubbles resonate particularly vigorously at the high frequencies used in ultrasound scans (they reflect these strong waves significantly more effectively than body tissues do). Since they are approximately the same size as red blood cells, they exhibit similar rheology in blood vessels and are used to measure blood flow in organs and tumors.

[0108] Conventionally, microbubbles have been used for drug delivery in the treatment of cancer. More recently, microbubbles have been used for molecular imaging and targeted gene delivery. It is believed that this is the first reporting of microbubbles being used to facilitate the release of biomarkers, genetic material, and/or non-genetic material across the blood brain barrier (BBB).

[0109] Microbubbles can be any microbubbles known in the art. For example, the microbubbles can be commercially available microbubbles, such as Definity® microbubbles or phase-change nanodroplets. For example, GE Healthcare makes Optison, a Food and Drug Administration (FDA)-approved microbubble. Optison has an albumin shell and octafluoropropane gas core. Lantheus makes Definity®, which is a FDA-approved microbubble. In Definity®, perflutren lipid microspheres are composed of octafluoropropane encapsulated in an outer lipid shell. Bracco makes Lumason (SonoVue in other countries) which is also FDA-approved. Lumason is a sulfur hexafluoride containing lipid-encapsulated microbubble.

[0110] Microbubbles can be used as described herein, but the use of microbubbles is not required to achieve the opening of the BBB and release of biomarkers or genetic information from a brain lesion or tumor.

Biomarker Detection, Genetic Testing, and Genetic Sequencing

[0111] Described herein are systems and methods for releasing biomarkers or genetic material from a brain into a biological sample for detection. Detection of these biomarkers can be used to diagnose or treat a subject or to formulate a personalized treatment plan. The presence or absence of such biomarkers or genetic material has implications on clinical decision-making in standard clinical care.

[0112] Described herein is the use of qPCR to detect mRNA, but any method of detecting genetic material or biomarkers known in the art can be used. It was also shown that brain-specific biomarkers were released following FUS to the brain (see e.g., Examples 3 and 4).

[0113] The disclosed systems and methods can be used to detect already known biomarkers for a disease or discover new biomarkers in subjects diagnosed with a disease.

[0114] Biomarkers, such as RNA or DNA, can have a short half-life. As such, the period of time for

the biological sample collection can be from about 1 minute to about 5 minutes or more after FUS is applied to the brain or a portion of the brain. For example, a sample can be collected before FUS, during FUS, or after FUS. As another example, the time after FUS application to collect a biological sample can be about 1 min, about 2 min; about 3 min; about 4 min; about 5 min; about 6 min; about 7 min; about 8 min; about 9 min; about 10 min; about 11 min; about 12 min; about 13 min; about 14 min; about 15 min; about 16 min; about 17 min; about 18 min; about 19 min; or about 20 min. A biological sample can be collected any time before the BBB closes. Recitation of each of these discrete values is understood to include ranges between each value. Recitation of each range is understood to include discrete values within the range.

[0115] A biological sample can be collected before FUS and after FUS to measure the biomarkers. Biomarkers can be detected to monitor a course of treatment or to monitor disease progression.

[0116] Widely-validated biomarkers that can be detected in a biological sample after FUS treatment can include (i) MGMT promoter methylation as a prognostic and predictive marker in glioblastoma; (ii) co-deletion of 1p and 19q differentiating oligodendrogliomas from astrocytomas; (iii) IDH1/2 mutations; or (iv) select pathway-associated mutations.

[0117] Biomarkers detected (with corresponding diseases presently known to be relevant for these biomarkers) can be: co-deletion 1p/19q (Oligodendrogliomas); MGMT promoter methylation (Glioblastomas, anaplastic astrocytomas); IDH 1/2 mutation (Oligo- and astrocytomas WHO grade II and III, secondary GBM); EGFRvIII (Glioblastomas); TERT mutation (Gliomas); ATRX mutations (Gliomas); TP53 mutations (Astrocytomas); BRAF V600 mutation (PXA, pilocytic astrocytomas, gangliogliomas); BRAF/KIAA1549 fusion (Pilocytic astrocytomas); H3.3 histones (Pediatric HGG); SHH pathway mutations (such as MYCN amplification, GLI2 amplification, or TP53 mutations) (Medulloblastomas); WNT pathway mutations (Medulloblastomas); or MYC (Medulloblastomas).

[0118] Detecting biomarkers of brain tumors in biological samples is well known in the art (see e.g., Staedtke et al. (2016) "Actionable Molecular Biomarkers in Primary Brain Tumors" Trends in Cancer 2(7) 338-349). Except as otherwise noted herein, therefore, the process of detecting biomarkers of the present disclosure can be carried out in accordance with such processes.

[0119] Detecting biomarkers (e.g., proteins, DNA, RNA, cell-free RNA, cell-free DNA) of neurodegenerative disease in biological samples is well known in the art (see e.g., Jeromin et al. (2017) Adv Neurobiol 15 491-528; Beach 217 Neurol Ther. 6(Supp 1)5-13). Except as otherwise noted herein, therefore, the process of detecting biomarkers of the present disclosure can be carried out in accordance with such processes.

[0120] As an example, the presently disclosed systems and methods can be used for detecting or discovering biomarkers for Alzheimer's disease (AD) (e.g., A β , tau, phosphorylated tau, NfL, or other neuronal proteins), Parkinson's disease (PD) (e.g., alpha-synuclein), or amyotrophic lateral sclerosis (ALS). In various examples, the presently disclosed systems and methods may be used for detecting or discovering biomarkers for neurodegenerative diseases. Non-limiting examples of neurodegenerative disease biomarkers include beta-amyloid (A β), A β 42, A β 42/A β 40, tau, phosphorylated tau (pTau), pTau181, pTau231, pTau217, N-terminal segment of tau, tau neurofibrillary tangles (NFT), neurofilament light chain (NfL), phosphorylated neurofilament heavy chain (pNfH), glial fibrillary acidic protein (GFAP), chitinase 3 like 1 (CHI3L1), interleukin 6 (IL-6), tumor necrosis factor alpha (TNF- α), α -synuclein, TAR DNA-binding protein 43 (TDP-43), neurogranin, SNAP-25, GAP-43, soluble platelet-derived growth factor receptor beta (sPDGFR- β), synaptic vesicle protein 2A (SV2A), neuromelanin, mutant HTT protein.

[0121] Detecting biomarkers of neurological or neurodegenerative diseases, disorders, or conditions in biological samples are well known in the art (see e.g., Smith 2017 Biomarkers on the brain: Putting biomarkers together for a better understanding of the nervous system, Science, special technology feature; Jain 2015 Biomarkers in Neurology, MedLink). Except as otherwise noted herein, therefore, the process of the present disclosure can be carried out in accordance with

such processes.

[0122] Detection and discovery of brain biomarkers can track disease progression over time and correlate with known clinical measures; detect the effect or efficacy of a drug; or serve as a surrogate endpoint in clinical trial.

Liquid Brain Biopsy

[0123] Described herein are methods and systems for blood-based liquid biopsies of the brain. The liquid brain biopsy can be used to detect genetic material from the brain from various central nervous system (CNS cancers or tumors or neurological or neurodegenerative diseases, disorders, or conditions). The disclosed methods and systems can reach lesions previously unreachable by traditional biopsy techniques. Furthermore, the disclosed systems and methods can be performed repeatedly for use in monitoring the lesion, treatment response, or treatment efficacy. The disclosed systems and methods can also be useful for diagnosing lesions or detecting biomarkers in areas of the brain or CNS that are typically not available for biopsy or surgery, such as the brain stem.

[0124] Previous methods for FUS on the brain focused on drug-delivery (“one-way trafficking”). Here, it was discovered that FUS-mediated BBB disruption enhances “two-way trafficking” between brain and blood.

[0125] Central nervous system (CNS) tumors are significant causes of cancer morbidity and mortality, especially in children and young adults where they account for ~20-30% of cancer deaths. Noninvasive neuroimaging modalities (e.g., magnetic resonance imaging (MRI), and computerized tomography (CT)) are used to evaluate tumor lesions, but observed changes, especially after treatment, can be difficult to interpret. Surgical resection or stereotactic biopsy is typically performed for histologic confirmation and increasingly for genetic profiling. However, tissue biopsy requires brain surgery and can be associated with adverse effects such as hemorrhage and infection. Repeated tumor biopsies that may be needed for tracking tumor evolution, treatment response, and tumor recurrence are often not feasible. Furthermore, tissue biopsies may be challenging when tumors are located at difficult locations or patients are too ill to tolerate invasive procedures.

[0126] Noninvasive blood-based liquid biopsies are a rapidly emerging strategy to provide genetic tumor profiling that can be used for treatment selection, treatment monitoring, residual disease detection, and asymptomatic individual screening. Some success has been achieved in integrating blood-based liquid biopsy into the routine clinical diagnostics. However, limited progress has been made for brain tumors. Although several groups have reported biomarker detection in the cerebrospinal fluid (CSF) of patients with brain tumors, obtaining CSF via lumbar puncture remains an invasive procedure and can be unsafe in some settings. Blood-based liquid biopsies are noninvasive, but there remain multiple hurdles for their application in the brain. First, the blood-brain barrier (BBB) restricts the release of large molecules such as DNA and RNA from the tumor to the peripheral circulation. Even though partial BBB disruption is a core feature of glioblastoma (the most common type of high-grade glioma) at the late stage of the tumors, the BBB remains intact in the early stage and the low-grade diffuse gliomas. Second, brain tumors, especially glioblastomas, are heterogeneous with spatially distinct molecular profiling. Localization of tumor areas that harbor specific mutations is not possible using conventional methods of liquid biopsy, as these methods are inherently spatially agnostic. Third, many tumor markers, such as some cell-free RNAs and DNAs, have short half-lives in blood. Detection of these biomarkers can be enhanced by stimulating their release from the tumor to the circulation and precisely controlling the blood-collection time to be shorter than their lifetimes in the blood. Therefore, the disclosed noninvasive, spatially resolved, and temporally controlled liquid biopsy technique is in great need to improve the clinical care of patients with any neurological or neurodegenerative disease, disorder, or condition (e.g., CNS tumors).

Central Nervous System (CNS) Cancer/Tumors

[0127] The presently disclosed methods can be used to detect or discover biomarkers for central

nervous system (CNS) cancers or tumors. CNS cancers can include a brain or spinal cord cancer or tumor.

[0128] The brain or spinal cord cancer or tumor detected can be acoustic neuroma; astrocytoma; atypical teratoid rhabdoid tumor (ATRT); brain stem glioma; chordoma; chondrosarcoma; choroid plexus; CNS lymphoma; craniopharyngioma; cysts; ependymoma; ganglioglioma; germ cell tumor; glioblastoma (GBM); glioma; hemangioma; juvenile pilocytic astrocytoma (JPA); lipoma; lymphoma; medulloblastoma; meningioma; metastatic brain tumor; neurilemmomas; neurofibroma; neuronal & mixed neuronal-glial tumors; non-hodgkin lymphoma; oligoastrocytoma; oligodendroglioma; optic nerve glioma; pineal tumor; pituitary tumor; primitive neuroectodermal (PNET); rhabdoid tumor; or schwannoma.

[0129] The astrocytoma can be a grade I pilocytic astrocytoma, a grade II—low-grade astrocytoma, a grade III anaplastic astrocytoma, a grade IV glioblastoma (GBM), or a juvenile pilocytic astrocytoma.

[0130] The glioma can be a brain stem glioma, an ependymoma, a mixed glioma, an optic nerve glioma, or a subependymoma.

[0131] Depending on the marker, treatment of CNS cancers or tumors can comprise the administration of PCV, temozolomide, IDH1/2inhibitors, Immunotherapeutic approaches (e.g., vaccines, CAR T-cells), BRAF inhibitors, MEK inhibitors, Epigenetic inhibitors, JMJD3 inhibitor, SMO inhibitors, reduced dose of RT, chemotherapy or combination, Gemcitabine and pemetrexed, or BET bromodomain inhibitors (see e.g., Staedtke et al. (2016) “Actionable Molecular Biomarkers in Primary Brain Tumors” Trends in Cancer 2(7) 338-349).

Neurological or Neurodegenerative Diseases, Disorders, or Conditions

[0132] These disclosed methods and system can also be used to detect or discover biomarkers for other neurological disease states. For example, the methods and systems can be used in subjects having or suspected of having a neurological disease, disorder, or condition such as Abulia; Agraphia; Alcoholism; Alexia; Alien hand syndrome; Allan-Herndon-Dudley syndrome; Alternating hemiplegia of childhood; Alzheimer's disease; Amaurosis fugax; Amnesia; Amyotrophic lateral sclerosis (ALS); Aneurysm; Angelman syndrome; Anosognosia; Aphasia; Apraxia; Arachnoiditis; Arnold-Chiari malformation; Asomatognosia; Asperger syndrome; Ataxia; Attention deficit hyperactivity disorder; ATR-16 syndrome; Auditory processing disorder; Autism spectrum; Behcet's disease; Bipolar disorder; Bell's palsy; Brachial plexus injury; Brain damage; Brain injury; Brain tumor; Brody myopathy; Canavan disease; Capgras delusion; Carpal tunnel syndrome; Causalgia; Central pain syndrome; Central pontine myelinolysis; Centronuclear myopathy; Cephalic disorder; Cerebral aneurysm; Cerebral arteriosclerosis; Cerebral atrophy; Cerebral autosomal dominant arteriopathy with subcortical infarcts and leukoencephalopathy (CADASIL); Cerebral dysgenesis-neuropathy-ichthyosis-keratoderma syndrome (CEDNIK syndrome); Cerebral gigantism; Cerebral palsy; Cerebral vasculitis; Cervical spinal stenosis; Charcot-Marie-Tooth disease; Chiari malformation; Chorea; Chronic fatigue syndrome; Chronic inflammatory demyelinating polyneuropathy (CIDP); Chronic pain; Cockayne syndrome; Coffin-Lowry syndrome; Coma; Complex regional pain syndrome; Compression neuropathy; Congenital facial diplegia; Corticobasal degeneration; Cranial arteritis; Craniosynostosis; Creutzfeldt-Jakob disease; Cumulative trauma disorders; Cushing's syndrome; Cyclothymic disorder; Cyclic Vomiting Syndrome (CVS); Cytomegalic inclusion body disease (CIBD); Cytomegalovirus Infection; Dandy-Walker syndrome; Dawson disease; De Morsier's syndrome; Dejerine-Klumpke palsy; Dejerine-Sottas disease; Delayed sleep phase syndrome; Dementia; Dermatomyositis; Developmental coordination disorder; Diabetic neuropathy; Diffuse sclerosis; Diplopia; Disorders of consciousness; Down syndrome; Dravet syndrome; Duchenne muscular dystrophy; Dysarthria; Dysautonomia; Dyscalculia; Dysgraphia; Dyskinesia; Dyslexia; Dystonia; Empty sella syndrome; Encephalitis; Encephalocele; Encephalotrigeminal angiomatosis; Encopresis; Enuresis; Epilepsy; Epilepsy-intellectual disability in females; Erb's palsy; Erythromelalgia; Essential tremor;

Exploding head syndrome; Fabry's disease; Fahr's syndrome; Fainting; Familial spastic paralysis; Febrile seizures; Fisher syndrome; Friedreich's ataxia; Fibromyalgia; Foville's syndrome; Fetal alcohol syndrome; Fragile X syndrome; Fragile X-associated tremor/ataxia syndrome (FXTAS); Gaucher's disease; Generalized epilepsy with febrile seizures plus; Gerstmann's syndrome; Giant cell arteritis; Giant cell inclusion disease; Globoid Cell Leukodystrophy; Gray matter heterotopia; Guillain-Barre syndrome; Generalized anxiety disorder; HTLV-1 associated myelopathy; Hallervorden-Spatz syndrome; Head injury; Headache; Hemifacial Spasm; Hereditary Spastic Paraplegia; Heredopathia atactica polyneuritiformis; Herpes zoster oticus; Herpes zoster; Hirayama syndrome; Hirschsprung's disease; Holmes-Adie syndrome; Holoprosencephaly; Huntington's disease; Hydranencephaly; Hydrocephalus; Hypercortisolism; Hypoxia; Immune-Mediated encephalomyelitis; Inclusion body myositis; Incontinentia pigmenti; Infantile Refsum disease; Infantile spasms; Inflammatory myopathy; Intracranial cyst; Intracranial hypertension; Isodicentric 15; Joubert syndrome; Karak syndrome; Kearns-Sayre syndrome; Kinsbourne syndrome; Kleine-Levin syndrome; Klippel Feil syndrome; Krabbe disease; Kufor-Rakeb syndrome; Lafora disease; Lambert-Eaton myasthenic syndrome; Landau-Kleffner syndrome; Lateral medullary (Wallenberg) syndrome; Learning disabilities; Leigh's disease; Lennox-Gastaut syndrome; Lesch-Nyhan syndrome; Leukodystrophy; Leukoencephalopathy with vanishing white matter; Lewy body dementia; Lissencephaly; Locked-in syndrome; Lou Gehrig's disease (See amyotrophic lateral sclerosis); Lumbar disc disease; Lumbar spinal stenosis; Lyme disease—Neurological Sequelae; Machado-Joseph disease (Spinocerebellar ataxia type 3); Macrencephaly; Macropsia; Mal de débarquement; Megalencephalic leukoencephalopathy with subcortical cysts; Megalencephaly; Melkersson-Rosenthal syndrome; Menieres disease; Meningitis; Menkes disease; Metachromatic leukodystrophy; Microcephaly; Micropsia; Migraine; Miller Fisher syndrome; Mini-stroke (transient ischemic attack); Misophonia; Mitochondrial myopathy; Mobius syndrome; Monomelic amyotrophy; Morvan syndrome; Motor Neuron Disease—see amyotrophic lateral sclerosis; Motor skills disorder; Moyamoya disease; Mucopolysaccharidoses; Multi-infarct dementia; Multifocal motor neuropathy; Multiple sclerosis; Multiple system atrophy; Muscular dystrophy; Myalgic encephalomyelitis; Myasthenia gravis; Myelinoclastic diffuse sclerosis; Myoclonic Encephalopathy of infants; Myoclonus; Myopathy; Myotubular myopathy; Myotonia congenita; Narcolepsy; Neuro-Behçet's disease; Neurofibromatosis; Neuroleptic malignant syndrome; Neurological manifestations of AIDS; Neurological sequelae of lupus; Neuromyotonia; Neuronal ceroid lipofuscinosis; Neuronal migration disorders; Neuropathy; Neurosis; Niemann-Pick disease; Non-24-hour sleep-wake disorder; Nonverbal learning disorder; O'Sullivan-McLeod syndrome; Occipital Neuralgia; Occult Spinal Dysraphism Sequence; Ohtahara syndrome; Olivopontocerebellar atrophy; Opsoclonus myoclonus syndrome; Optic neuritis; Orthostatic Hypotension; Otosclerosis; Overuse syndrome; Palinopsia; Paresthesia; Parkinson's disease; Paramyotonia congenita; Paraneoplastic diseases; Paroxysmal attacks; Parry-Romberg syndrome; PANDAS; Pelizaeus-Merzbacher disease; Periodic paralyses; Peripheral neuropathy; Pervasive developmental disorders; Phantom limb/Phantom pain; Photic sneeze reflex; Phytanic acid storage disease; Pick's disease; Pinched nerve; Pituitary tumors; PMG; Polyneuropathy; Polio; Polymicrogyria; Polymyositis; Porencephaly; Post-polio syndrome; Postherpetic neuralgia (PHN); Postural hypotension; Prader-Willi syndrome; Primary lateral sclerosis; Prion diseases; Progressive hemifacial atrophy; Progressive multifocal leukoencephalopathy; Progressive supranuclear palsy; Prosopagnosia; Pseudotumor cerebri; Quadrantanopia; Quadriplegia; Rabies; Radiculopathy; Ramsay Hunt syndrome type I; Ramsay Hunt syndrome type II; Ramsay Hunt syndrome type III—see Ramsay-Hunt syndrome; Rasmussen encephalitis; Reflex neurovascular dystrophy; Refsum disease; REM sleep behavior disorder; Repetitive stress injury; Restless legs syndrome; Retrovirus-associated myelopathy; Rett syndrome; Reye's syndrome; Rhythmic Movement Disorder; Romberg syndrome; Saint Vitus dance; Sandhoff disease; Schilder's disease (two distinct conditions); Schizencephaly; Sensory processing disorder; Septo-optic dysplasia; Shaken baby syndrome;

Shingles; Shy-Drager syndrome; Sjögren's syndrome; Sleep apnea; Sleeping sickness; Snatiation; Sotos syndrome; Spasticity; Spina bifida; Spinal cord injury; Spinal cord tumors; Spinal muscular atrophy; Spinal and bulbar muscular atrophy; Spinocerebellar ataxia; Split-brain; Steele-Richardson-Olszewski syndrome; Stiff-person syndrome; Stroke; Sturge-Weber syndrome; Stuttering; Subacute sclerosing panencephalitis; Subcortical arteriosclerotic encephalopathy; Superficial siderosis; Sydenham's chorea; Syncope; Synesthesia; Syringomyelia; Tarsal tunnel syndrome; Tardive dyskinesia; Tardive dysphrenia; Tarlov cyst; Tay-Sachs disease; Temporal arteritis; Temporal lobe epilepsy; Tetanus; Tethered spinal cord syndrome; Thomsen disease; Thoracic outlet syndrome; Tic Douloureux; Todd's paralysis; Tourette syndrome; Toxic encephalopathy; Transient ischemic attack; Transmissible spongiform encephalopathies; Transverse myelitis; Traumatic brain injury; Tremor; Trichotillomania; Trigeminal neuralgia; Tropical spastic paraparesis; Trypanosomiasis; Tuberous sclerosis; 22q13 deletion syndrome; Unverricht-Lundborg disease; Vestibular schwannoma (Acoustic neuroma); Von Hippel-Lindau disease (VHL); Viliuisk Encephalomyelitis (VE); Wallenberg's syndrome; West syndrome; Whiplash; Williams syndrome; Wilson's disease; Y-Linked Hearing Impairment; or Zellweger syndrome.

[0133] For example, the methods and systems can be used in subjects having or suspected of having a neurological disease, disorder, or condition comprising a neurodegenerative disease, disorder, or condition. As an example, a neurodegenerative disease, disorder or condition can be Alzheimer's disease, amyotrophic lateral sclerosis (ALS), Alexander disease, Alpers' disease, Alpers-Huttenlocher syndrome, alpha-methylacyl-CoA racemase deficiency, Andermann syndrome, Arts syndrome, ataxia neuropathy spectrum, ataxia (E.g., with oculomotor apraxia, autosomal dominant cerebellar ataxia, deafness, and narcolepsy), autosomal recessive spastic ataxia of Charlevoix-Saguenay, Batten disease, beta-propeller protein-associated neurodegeneration, Cerebro-Oculo-Facio-Skeletal Syndrome (COFS), Corticobasal Degeneration, CLN1 disease, CLN10 disease, CLN2 disease, CLN3 disease, CLN4 disease, CLN6 disease, CLN7 disease, CLN8 disease, cognitive dysfunction, congenital insensitivity to pain with anhidrosis, dementia, familial encephalopathy with neuroserpin inclusion bodies, familial British dementia, familial Danish dementia, fatty acid hydroxylase-associated neurodegeneration, Gerstmann-Straussler-Scheinker Disease, GM2-gangliosidosis (e.g., AB variant), HMSN type 7 (e.g., with retinitis pigmentosa), Huntington's disease, infantile neuroaxonal dystrophy, infantile-onset ascending hereditary spastic paralysis, Huntington's disease (HD), infantile-onset spinocerebellar ataxia, juvenile primary lateral sclerosis, Kennedy's disease, Kuru, Leigh's Disease, Marinesco-Sjögren syndrome, Mild Cognitive Impairment (MCI), mitochondrial membrane protein-associated neurodegeneration, Motor neuron disease, Monomelic Amyotrophy, Motor neuron diseases (MND), Multiple System Atrophy, Multiple System Atrophy with Orthostatic Hypotension (Shy-Drager Syndrome), multiple sclerosis, multiple system atrophy, neurodegeneration in Down's syndrome (NDS), neurodegeneration of aging, Neurodegeneration with brain iron accumulation, neuromyelitis optica, pantothenate kinase-associated neurodegeneration, Opsoclonus Myoclonus, prion disease, Progressive Multifocal Leukoencephalopathy, Parkinson's disease (PD), PD-related disorders, polycystic lipomembranous osteodysplasia with sclerosing leukoencephalopathy, prion disease, progressive external ophthalmoplegia, riboflavin transporter deficiency neuropathy, Sandhoff disease, Spinal muscular atrophy (SMA), Spinocerebellar ataxia (SCA), Striatonigral degeneration, Transmissible Spongiform Encephalopathies (Prion Diseases), or Wallerian-like degeneration.

[0134] In some examples, the systems and methods described herein can be used to target a region of a brain of a patient with a neurologic disease with no radiographically visible lesion. For example, the systems and methods described herein can be used to target a region of a brain of a patient with psychiatric diagnoses, depression, bipolar disorder, schizophrenia, neurodevelopmental disorders, autism, ADHD, neurodegenerative disorders, Alzheimer's, frontotemporal dementia, Lewy body disease, Parkinsons.

[0135] In some examples, the systems and methods described herein may be used in combination

with diagnostic imaging to improve the diagnosis of a particular disease or condition. Non-limiting examples of diagnostic imaging include MRI, PET, SPECT, and ultrasound. For example, the diagnosis of a neurodegenerative disease in a patient may be improved with the use of FUS-LBx in combination with diagnostic imaging.

Diagnosis of Brain Cancer in Liquid Biopsy Samples

[0136] Further provided herein is a method for diagnosing brain cancer utilizing the systems and methods described herein. For example, the method for diagnosing brain cancer can include obtaining a biological sample after opening the BBB and releasing one or more biomarkers, as described herein. Patients with brain cancer have circulating tumor DNA (ctDNA). ctDNA is a type of cell free DNA (cfDNA). ctDNA exhibits different properties from healthy patient cfDNA. For example, cfDNA which includes ctDNA has generally different fragmentomic features from healthy patient cfDNA (e.g., different ratio of short fragments to long fragments, different median fragment length, and different coefficient of variation (CV) of the fragment length distribution at the transcription start sites (TSSs)). In some examples, the brain cancer can be glioma, glioblastoma, or other types of brain cancer.

[0137] Fragmentomic features of cfDNA can be determined using high-throughput sequencing methods followed by bioinformatic analysis to extract fragment-level data.

[0138] FIG. 25 illustrates a flow chart of an exemplary method for diagnosing brain cancer. The method **100** can be used to diagnose brain cancer based on fragmentomic features of cfDNA released across a blood brain barrier of a patient. At block **102**, the method **100** can begin by applying a focused ultrasound to a brain or brain region of a patient. The focused ultrasound can be operable to disrupt a blood brain barrier of the patient and release cfDNA across the blood brain barrier of the patient. In some examples, the focused ultrasound can be applied to the brain or brain region using the systems and methods described herein. For example, the focused ultrasound can be applied for a period of time sufficient to disrupt the blood brain barrier. In some examples, the period of time can be about 1 minute to about 4 minutes. In some examples, the period of time can be about 1 minute to about 30 minutes. In some examples, the period of time can be greater than 30 minutes. The focused ultrasound can be applied at an acoustic pressure sufficient to disrupt the blood brain barrier and release the cfDNA across the blood brain barrier. In some examples, the acoustic pressure can be about 0.1 MPa to about 10 MPa.

[0139] In some examples, the focused ultrasound is applied to a brain region that includes a suspected tumor. The focused ultrasound can be operable to release cfDNA from the suspected tumor, which can include ctDNA. In some examples, the suspected tumor can be located (e.g., identified) before the method **100** begins. In some examples, the suspected tumor can be identified by an MRI machine, CT scanning, and/or ultrasound imaging.

[0140] At block **104**, the method **100** can include obtaining a biological sample from the patient. In some examples, the biological sample can be blood. In some examples, the biological sample can be collected using a syringe. In some examples, the biological sample can be collected using other methods for collecting biological samples. The biological sample can include the cfDNA released across the blood brain barrier. For example, the biological sample can be collected at a collection time after the focused ultrasound was applied. In some examples, the collection time can be about 1 minute to about 30 minutes after the focused ultrasound was applied. In some examples, the collection time can be about 1 minute to about 5 minutes, about 5 minutes to about 10 minutes, about 10 minutes to about 15 minutes, about 15 minutes to about 20 minutes, about 20 minutes to about 25 minutes, or about 25 minutes to about 30 minutes after the focused ultrasound was applied. In some examples, the collection time allows for the cfDNA within the blood brain barrier to enter the bloodstream before collecting the biological sample.

[0141] At block **106**, the method **100** can include measuring one or more diagnostic fragmentomic features of the cfDNA in the biological sample. In some examples, the one or more diagnostic fragmentomic features of the cfDNA can include one or more of a ratio of short fragments (e.g.,

less than 120 base pairs (bp)) to long fragments (e.g., 140 bp to 250 bp), a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes.

[0142] At block **108**, the method **100** can include comparing the one or more diagnostic fragmentomic features to one or more thresholds. In some examples, the one or more thresholds can include a percentage of increase or decrease in the one or more diagnostic fragmentomic features to one or more baseline fragmentomic features. For example, prior to applying the focused ultrasound, the method **100** can include obtaining a baseline biological sample. The baseline biological sample does not include the cfDNA from within the blood brain barrier. One or more baseline fragmentomic features can be measured from the baseline biological sample. The one or more baseline fragmentomic features can be the same fragmentomic features as the one or more diagnostic fragmentomic features (e.g., a ratio of short fragments (e.g., less than 120 base pairs (bp)) to long fragments (e.g., 140 bp to 250 bp), a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes). In some examples, cfDNA that is more fragmented after focused ultrasound application (e.g., after cfDNA is released from the blood brain barrier) is indicative of the presence of ctDNA, and thereby indicative of brain cancer.

[0143] In some examples, the percentage increase or decrease between the one or more baseline fragmentomic features and the one or more diagnostic fragmentomic features can be about 10% to about 20% or more. For example, an increase of about 10% or more in the ratio of short fragments to long fragments between the baseline biological sample (e.g., taken before focused ultrasound releases cfDNA through the blood brain barrier) and the diagnostic biological sample (e.g., taken after applying the focused ultrasound and releasing cfDNA through the blood brain barrier) can be indicative that the patient has ctDNA within the brain. For example, ctDNA has smaller fragments than healthy cfDNA. The presence of ctDNA can be indicative of brain cancer. Similarly, a decrease of about 10% to about 20% or more in median fragment length between the baseline biological sample and the diagnostic biological sample can be indicative that the patient has ctDNA within the brain. Similarly, an increase of about 10% to about 20% or more in the coefficient of variation of the fragment length distribution at the transcription start sites (TSSs) of all protein-coding genes between the baseline biological sample and the diagnostic biological sample can be indicative that the patient has ctDNA within the brain.

[0144] In other examples, the one or more thresholds can be empirically determined thresholds. For example, one or more thresholds for the fragmentomic features can be determined from a population of healthy patients and/or a population of patients with brain cancer. For example, fragmentomic features (e.g., short to long fragment ratio, median fragment length, and/or coefficient of variation of the fragment length distribution at the transcription start sites) can be determined for healthy patients and for patients with brain cancer. Threshold values can be determined based on the fragmentomic features in healthy patients and/or patients with brain cancer. For example, a threshold of $\pm 10\%$ can be set from an average fragmentomic feature value of healthy patients. Beside single-feature thresholds, multivariate thresholding and machine learning models could be employed. These approaches integrate multiple fragmentomic features into a composite predictive model that can distinguish between healthy individuals and patients with brain cancer more robustly than univariate methods.

[0145] At block **110**, the method **100** can include diagnosing the patient with brain cancer when the one or more diagnostic fragmentomic features exceed the one or more thresholds. As described herein, the values of the diagnostic fragmentomic features can be indicative of the presence or absence ctDNA in the brain (e.g., after cfDNA is released through the blood brain barrier and collected in the biological sample). The presence of ctDNA is indicative that the patient has brain cancer. For example, the one or more diagnostic fragmentomic features will exhibit shorter fragment lengths (and related values as described herein) if ctDNA is present in the cfDNA,

thereby indicating that the patient has brain cancer.

[0146] In some examples, the method **100** can further include assessing an effectiveness of the disruption of the blood brain barrier or release of cfDNA by measuring imaging contrast before and after applying the focused ultrasound. Assessing effectiveness can be conducted using an MRI machine, CT scanner, or ultrasound imaging system. The imaging contrast can include MRI contrast, CT contrast, and/or ultrasound contrast. In some examples, an increase in imaging contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

[0147] In some examples, the method **100** can further include administering microbubbles to the patient. The microbubbles can be administered in an amount sufficient to disrupt the blood brain barrier when applying focused ultrasound, as described herein.

[0148] In some examples, the method **100** can include detecting, via an acoustic sensor, an acoustic signal, as described herein. In some examples, the acoustic sensor can be operable to measure and/or monitor cavitation acoustic emissions.

[0149] Definitions and methods described herein are provided to better define the present disclosure and to guide those of ordinary skill in the art in the practice of the present disclosure. Unless otherwise noted, terms are to be understood according to conventional usage by those of ordinary skill in the relevant art.

[0150] In some embodiments, numbers expressing quantities of ingredients, properties such as molecular weight, reaction conditions, and so forth, used to describe and claim certain embodiments of the present disclosure are to be understood as being modified in some instances by the term “about.” In some embodiments, the term “about” is used to indicate that a value includes the standard deviation of the mean for the device or method being employed to determine the value. In some embodiments, the numerical parameters set forth in the written description and attached claims are approximations that can vary depending upon the desired properties sought to be obtained by a particular embodiment. In some embodiments, the numerical parameters should be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of some embodiments of the present disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as practicable. The numerical values presented in some embodiments of the present disclosure may contain certain errors necessarily resulting from the standard deviation found in their respective testing measurements. The recitation of ranges of values herein is merely intended to serve as a shorthand method of referring individually to each separate value falling within the range. Unless otherwise indicated herein, each individual value is incorporated into the specification as if it were individually recited herein.

[0151] In some embodiments, the terms “a” and “an” and “the” and similar references used in the context of describing a particular embodiment (especially in the context of certain of the following claims) can be construed to cover both the singular and the plural, unless specifically noted otherwise. In some embodiments, the term “or” as used herein, including the claims, is used to mean “and/or” unless explicitly indicated to refer to alternatives only or the alternatives are mutually exclusive.

[0152] The terms “comprise,” “have” and “include” are open-ended linking verbs. Any forms or tenses of one or more of these verbs, such as “comprises,” “comprising,” “has,” “having,” “includes” and “including,” are also open-ended. For example, any method that “comprises,” “has” or “includes” one or more steps is not limited to possessing only those one or more steps and can also cover other unlisted steps. Similarly, any composition or device that “comprises,” “has” or “includes” one or more features is not limited to possessing only those one or more features and can cover other unlisted features.

[0153] As used herein, the terms “sonobiopsy”, “liquid biopsy”, and “FUS-LBx” may be used interchangeably to mean the use of focused ultrasound to non-invasively release biomarkers from a

brain or a brain region across the blood brain barrier (BBB) of a subject.

[0154] As used herein, “genetic material” may include proteins, extracellular vesicles, cell-free RNA, cell-free DNA, mRNA, circulating tumor DNA (ctDNA), DNA methylation, or plasma DNA concentration. For example, genetic material may include any biomarkers disclosed herein.

[0155] All methods described herein can be performed in any suitable order unless otherwise indicated herein or otherwise clearly contradicted by context. The use of any and all examples, or exemplary language (e.g. “such as”) provided with respect to certain embodiments herein is intended merely to better illuminate the present disclosure and does not pose a limitation on the scope of the present disclosure otherwise claimed. No language in the specification should be construed as indicating any non-claimed element essential to the practice of the present disclosure.

[0156] Groupings of alternative elements or embodiments of the present disclosure disclosed herein are not to be construed as limitations. Each group member can be referred to and claimed individually or in any combination with other members of the group or other elements found herein. One or more members of a group can be included in, or deleted from, a group for reasons of convenience or patentability. When any such inclusion or deletion occurs, the specification is herein deemed to contain the group as modified thus fulfilling the written description of all Markush groups used in the appended claims.

[0157] All publications, patents, patent applications, and other references cited in this application are incorporated herein by reference in their entirety for all purposes to the same extent as if each individual publication, patent, patent application or other reference was specifically and individually indicated to be incorporated by reference in its entirety for all purposes. Citation of a reference herein shall not be construed as an admission that such is prior art to the present disclosure.

[0158] Having described the present disclosure in detail, it will be apparent that modifications, variations, and equivalent embodiments are possible without departing the scope of the present disclosure defined in the appended claims. Furthermore, it should be appreciated that all examples in the present disclosure are provided as non-limiting examples.

EXAMPLES

[0159] The following non-limiting examples are provided to further illustrate the present disclosure. It should be appreciated by those of skill in the art that the techniques disclosed in the examples that follow represent approaches the inventors have found function well in the practice of the present disclosure, and thus can be considered to constitute examples of modes for its practice. However, those of skill in the art should, in light of the present disclosure, appreciate that many changes can be made in the specific embodiments that are disclosed and still obtain a like or similar result without departing from the spirit and scope of the present disclosure.

Example 1: Focused Ultrasound-Enabled Brain Tumor Liquid Biopsy

[0160] The following example describes the detection of a glioblastoma biomarker (mRNA) in an animal model.

[0161] Although blood-based liquid biopsies have emerged as a promising non-invasive method to detect biomarkers in various cancers, limited progress has been made for brain tumors. One major obstacle is the blood-brain barrier (BBB), which hinders efficient passage of tumor biomarkers into the peripheral circulation. The objective of this study was to determine whether FUS in combination with microbubbles can enhance the release of biomarkers from the brain tumor to the blood circulation. Two glioblastoma tumor models (U87 and GL261), developed by intracranial injection of respective enhanced green fluorescent protein (eGFP)-transduced glioblastoma cells, were treated by FUS in the presence of systemically injected microbubbles. Effect of FUS on plasma eGFP mRNA levels was determined using quantitative polymerase chain reaction. eGFP mRNA were only detectable in the FUS-treated U87 mice and undetectable in the untreated U87 mice (maximum cycle number set to 40). This finding was replicated in GL261 mice across three different acoustic pressures. The circulating levels of eGFP mRNA were 1,500-4,800 fold higher in

the FUS-treated GL261 mice than that of the untreated mice for the three acoustic pressures. This study demonstrated the feasibility of FUS-enabled brain tumor liquid biopsies in two different murine glioma models across different acoustic pressures.

Introduction

[0162] Mutations in the DNA, changes in epigenomic makeup, and variations in gene expression associated with brain tumors can inform clinical practice by providing invaluable information for diagnosis, prognostication, disease monitoring, and development of personalized treatment strategies.^{sup.1} Such molecular biomarkers, which can be examined in surgical resection or biopsy specimens, are becoming an integral component of clinical practice.^{sup.2} However, direct surgical tissue biopsy to determine tumor molecular profiles is associated with potential complications such as hemorrhage and infection.^{sup.3} Furthermore, repeated tissue biopsies using surgical interventions to assess treatment response and recurrence may not be feasible given the increased risk for complications and morbidity, especially for brain tumors. Liquid biopsy offers a noninvasive approach for detecting circulating molecular biomarkers.

[0163] Although liquid biopsy has been established in clinical care of patients with various cancer types.^{sup.4}, limited progress has been made for brain tumors.^{sup.2} For brain tumor liquid biopsy, the major challenge is the hindrance of tumor biomarker release into the bloodstream by the blood-brain barrier (BBB).^{sup.5} Even though the BBB is partially disrupted in the core part of glioblastoma (the most common type of high-grade glioma), it remains intact in large parts of glioblastoma and lower grade diffuse gliomas, which may prevent the efficient passage of biomarkers into the blood circulation. In line with this, circulating tumor DNA (ctDNA) was reported to be detectable in a small fraction of patients with advanced gliomas (<10%) as compared to patients with other solid tumors.^{sup.6} It has also been shown that patients with high-grade glioma have a significantly higher plasma DNA concentration than patients with low-grade gliomas. These suggest that the increased permeability of BBB associated with the progression of gliomas is correlated with the release of biomarkers from the tumor to the blood.^{sup.7} It was also found that although D-2-hydroxyglutarate (D2HG) levels have been used in the clinic for the diagnosis and monitoring of patients with IDH1/2-mutant malignancies, D2HG plasma levels in patients with IDH2/2-mutant gliomas are within the normal range, suggesting that the BBB prevents D2HG from entering the blood circulation.^{sup.8} Therefore, approaches that can non-invasively enhance the release of biomarkers from the brain tumors to the blood circulation would be of significant clinical relevance.

[0164] Focused ultrasound (FUS) offers noninvasive and spatially-localized biomarker release into the blood stream.^{sup.39} The initial concept for blood biomarker amplification and localization using ultrasound was proposed in a study published in 2009, which demonstrated the feasibility to enhance the release of protein biomarkers from a colon cancer cell line in both ex vivo cell cultures and an in vivo mouse tumor model.^{sup.10} Building on this initial work, in vitro cell culture studies showed the feasibility of enhancing mRNA biomarker release from a breast cancer cell line using microbubble-enhanced ultrasound.^{sup.11}, and the feasibility of releasing a combination of ovarian cancer biomarkers using ultrasound.^{sup.12} Two recent in vivo studies further demonstrated that ultrasound-mediated release of biomarkers into the bloodstream is a promising approach for detecting tumor biomarkers via blood sampling.^{sup.10,13,14} In one of the studies, a chicken embryo tumor model was used to show the feasibility of amplifying the release of extracellular vesicles using high-intensity focused ultrasound (HIFU) in combination with phase-change nanodroplets which changed to microbubbles upon HIFU sonication.^{sup.14} In the other study, pulsed HIFU with high acoustic pressures was used to induce histotripsy (i.e., a technique for mechanical tissue fractionation) in a rat model of prostate cancer, and this enhanced release of cell-free tumor microRNA into the blood circulation.^{sup.13} Although promising, these findings cannot be readily extended to applications in the brain, given challenges inherent to the brain: first, delivery of acoustic energy to the brain is impeded by attenuation and distortion of acoustic waves

by the skull; second, biomarker release from the brain is inherently limited by the presence of the BBB.

[0165] FUS in combination with microbubbles has been studied extensively for inducing BBB opening for noninvasive and localized delivery of drugs in the blood circulation to the brain parenchyma.^{sup.15-17} Many studies have been performed to optimize the treatment parameters.^{sup.18-20} and evaluate the short-term and long-term safety profiles.^{sup.21-24} Ongoing clinical trials are evaluating the feasibility and safety of FUS-induced BBB opening in patients with glioblastoma and Alzheimer's disease.^{sup.25,26} Although all previous studies focused on exploring the utility of FUS-induced BBB as a means of targeted delivery of circulating therapeutics, here, it was hypothesized that FUS-mediated BBB disruption could be viewed as a tool for enhancing “two-way trafficking” between brain and blood. While circulating molecules can be allowed to enter the brain using FUS-mediated BBB disruption, brain biomarkers (e.g., tumor markers) can also be released into the blood circulation for liquid biopsies. Here is disclosed the development of an FUS-enabled brain tumor liquid biopsy technique, which uses FUS in combination with microbubbles to enhance the release of biomarkers from brain tumors into the blood circulation for liquid biopsies.

[0166] The feasibility of using FUS in combination with microbubbles was demonstrated for the local release of mRNA from glioblastoma tumors in mice into the bloodstream for liquid biopsies. Glioblastoma was selected as the tumor model because it is the most frequent type of primary brain cancer in adults and associated with a dismal prognosis.^{sup.27} The biomarker used was enhanced green fluorescent protein (eGFP) mRNA, which was highly specific to the tumor models used in this study, as the tumor models were established by the direct injection of eGFP-luciferase-transduced glioblastoma cells into the mouse brain. Both human glioma U87 cells and murine glioma GL261 cells were used to develop two mouse models of glioblastoma.

Results

FUS-Enabled Liquid Biopsy in an Orthotopic Human Glioma Xenograft Model

[0167] An orthotopic mouse model developed by implantation of eGFP-luciferase-transduced human glioma cells (U87) into the brains of nude mice was treated by an ultrasound imaging-guided FUS system (see e.g., FIG. 1A). Local growth of the tumors within the brain was assessed by monitoring luciferase activity using bioluminescent imaging (BLI) and verified using fluorescence imaging of the ex vivo brain slices (see e.g., FIG. 1B). The focus of the FUS transducer (frequency=1.5 MHz) was targeted at the tumor based on the tumor location identified by the BLI. The acoustic pressure of the FUS pulses was 3.28 MPa and other parameters were similar to those used for the BBB disruption (duty cycle of 1%, pulse repetition frequency of 1 Hz, and exposure duration of 2 min) (see e.g., FIG. 1E).^{sup.20-24} The experimental timeline is shown in FIG. 1F. For the U87 mice, terminal (non-survival) blood collection via cardiac puncture was performed immediately (~4 minutes) after FUS treatment. Such a short interval was selected in reference to a recent observation showing shorter collection time after FUS treatment was associated with higher RNA yield.^{sup.13}

[0168] Quantitative polymerase chain reaction (qPCR) was performed to determine relative circulating levels of eGFP mRNA (target biomarker) normalized to 5.8S rRNA (internal control) in blood serum. Two pairs of PCR primers were used for the quantification of eGFP, namely eGFP A and eGFP B (TABLE 1). PCR products for the two eGFP primer pairs were undetectable in the control mice without FUS treatment with the qPCR maximum cycle number set to 40 (TABLE 2). Amplification curves of circulating eGFP mRNA in control and treated mice for the two primer pairs are shown in FIG. 2A. Quantification of the eGFP mRNA in blood collected from U87 mice found the average±standard deviation of $\Delta C_{sub.T}$ for eGFP A and eGFP B were both 30.7 ± 0.8 for the control group without FUS treatment. For the FUS-treated mice, the average±standard deviation of $\Delta C_{sub.T}$ for eGFP A and eGFP B were 16.6 ± 5.2 and 19.4 ± 5.4 , respectively (TABLE 2).

TABLE-US-00001 TABLE 1 Forward and reverse primers used in qPCR for eGFP mRNA and 5.8s rRNA. Two primers were used for eGFP quantification, called eGFP A and eGFP B. 5.8s rRNA was used as an internal control. Primer Forward Reverse eGFP A AGAACGGCATCAAGGTGAA TGCTCAGGTAGTGGTTG C (SEQ ID NO: 1) TCG (SEQ ID NO: 4) eGFP B TATATCATGGCCGACAAGC ACTGGGTGCTCAGGTAG A (SEQ ID NO: 2) TGG (SEQ ID NO: 5) 5.8s rRNA GACTCTTAGCGGTGGATCA CGTTCTTCATCGACGCA CT (SEQ ID NO: 3) CGA (SEQ ID NO: 6)

TABLE-US-00002 TABLE 2 Summary of normalized cycle threshold, $\Delta C_{sub.T}$, for eGFP A and eGFP B in the U87 control mice (C1-C3; n = 3) and treated mice (T1-T6; n = 6). Mice eGFP A eGFP B Identifier $\Delta C_{sub.T}$ $\Delta C_{sub.T}$ Control Mice C1 29.9 29.9 C2 30.9 30.9 C3 31.4 31.4 Treated Mice T1 14.8 25.9 T2 19.6 19.3 T3 25.1 25.1 T4 12.8 13.6 T5 16.4 19.1 T6 10.8 13.3 [0169] 2.sup. $-\Delta C_{sub.T}$ was calculated to compare the relative gene expression levels of the FUS-treated mice and the control mice (see e.g., FIG. 2B and FIG. 2C). For both eGFP primer pairs, circulating mRNA levels of eGFP were significantly higher in the FUS-treated group compared with the untreated control group (eGFP A: p=0.01; eGFP B: p=0.01; one-tailed non-parametric Mann Whitney U Test).

FUS-Enabled Liquid Biopsy in an Orthotopic Murine Xenograft Glioma Model

[0170] The second orthotopic glioma model was developed by direct implantation of murine glioma cells (GL261) into the Swiss mice. A magnetic resonance-guided FUS (MRgFUS) system was used for the FUS sonication to achieve accurate tumor targeting (see e.g., FIG. 1C). The MRgFUS system, which was operated at 1.44 MHz, was targeted at the center of the tumor. Three groups of GL261 mice were treated by FUS with acoustic pressures of 1.48 MPa, 2.10 MPa, and 3.34 MPa, respectively. All other parameters were kept the same as those used in the treatment of U87 mice. Contrast-enhanced MR images were acquired before FUS sonication to identify the location of the tumor for FUS targeting and after FUS sonication to verify accurate tumor targeting by the FUS (see e.g., FIG. 1D).

[0171] Quantification of the eGFP mRNA in the blood collected from GL261 mice found the average $\pm$ standard deviation of $\Delta C_{sub.T}$ for eGFP A and eGFP B were both 23.6 ± 0.2 for the control group. The mean $\pm$ standard deviation of $\Delta C_{sub.T}$ for eGFP A was 11.6 ± 0.1 for the 1.48 MPa group, 12.5 ± 0.4 for the 2.01 MPa group, and 12.2 ± 0.1 for the 3.34 MPa group (TABLE 3). The mean $\pm$ standard deviation of $\Delta C_{sub.T}$ for eGFP B were 12.2 ± 0.3 , 13.7 ± 0.7 , and 13.0 ± 0.2 for the three groups, respectively (TABLE 3).

TABLE-US-00003 TABLE 3 Summary of normalized cycle threshold, $\Delta C_{sub.T}$, for eGFP A and eGFP B in the GL261 control mice (C1-C3; n = 3) and treated mice with different acoustic pressures. Mice eGFP A eGFP B Identifier $\Delta C_{sub.T}$ $\Delta C_{sub.T}$ Control Mice C1 23.6 23.6 C2 23.9 23.8 C3 23.4 23.4 Treated Mice 1.48 MPa 11.6 12.4 1.48 MPa 11.5 11.8 1.48 MPa 11.7 12.3 2.10 MPa 12.5 13.9 2.10 MPa 12.0 12.8 2.10 MPa 12.9 14.5 3.34 MPa 12.3 13.2 3.34 MPa 12.2 13.1 3.34 MPa 12.1 12.8

[0172] Regardless of acoustic pressure, circulating eGFP levels were significantly higher (1,500-4,800 fold higher) in the FUS-treated groups (n=9 in total, eGFP A: p=0.0045; eGFP B: p=0.0045; one-tailed Mann Whitney U Test) relative to the control group (n=3, FIG. 3A-B). The expression levels of mice treated at the lowest pressure (n=3; 1.48 MPa) were significantly higher than those of the other two groups for eGFP A and eGFP B (n=6; eGFP A: 1.7 fold increase in average, p=0.012; eGFP B: 2.2 fold increase in average; p=0.012; one-tailed Mann Whitney U Test). This finding suggests that the relatively lower pressure (1.48 MPa) was more efficient in releasing eGFP mRNA from the tumor than the relatively higher pressures (2.10 MPa and 3.34 MPa). Additionally, these MRI data found that the MR contrast enhancement ratios (calculated by the intensities of the MR images acquired after FUS treatment divided by the intensities of the images obtained before FUS treatment) were not significantly different among the three pressure groups.

Histological Examination

[0173] H&E staining of the GL261 mouse brains found red blood cell extravasation in all mice treated with FUS (see e.g., FIG. 4A-D). More severe hemorrhage was observed in brain slices obtained from mice treated with the relatively higher pressures (2.10 MPa and 3.34 MPa) than mice treated with the relatively lower pressure (1.48 MPa). Vascular damage was expected as the acoustic pressures used in this study were higher than the pressures normally used for the BBB opening without causing vascular damage. Of note, hemorrhage was not observed in the U87 mice treated by FUS at 3.28 MPa. The short interval between FUS sonication and animal scarification in the U87 mice (about 4 minutes vs. 20 minutes in GL261 mice) may have precluded the appearance of red blood cells in the brain slices even in the presence of tissue damage.

DISCUSSION

[0174] This study demonstrated the use of FUS-enabled brain liquid biopsy using two glioma mouse models. FUS in combination with microbubbles achieved noninvasive and spatially-localized biomarker release from the brain tumor into the bloodstream. The noninvasive nature of the disclosed FUS technique is especially advantageous over conventional neurosurgical tissue biopsies. Moreover, the technique presents a unique advantage in the assessment of spatially heterogeneous tumors. Tumor heterogeneity, which is a hallmark of glioblastoma^{sup.28}, poses a significant challenge to cancer biomarker research^{sup.29}. FUS can precisely target different locations of the tumor, thereby causing biomarkers to be released in a spatially-localized manner. By targeting multiple tumor regions in a single FUS session, the technique can be used to capture and analyze spatially heterogeneous biomarkers in a single liquid biopsy sample. Another potential functionality would be to perform multiple FUS sessions, each followed by a liquid biopsy, in order to detect specific biomarkers for each spatial location of the tumor to better understand the spatial heterogeneity of the tumor.

[0175] Both ultrasound-imaging guided FUS and MRgFUS treatment yielded substantial increase in eGFP RNA levels. However, the standard deviations of the qPCR measurement results for FUS-treated GL261 mice were lower than those of the FUS-treated U87 mice (see e.g., FIG. 2A-2C, FIG. 3A-B). This decrease in variations of the experimental results was associated with the use of the MRgFUS system, which improved the tumor targeting accuracy compared with the ultrasound imaging-guided FUS system.

[0176] Although damage was observed, the FUS-enabled brain liquid biopsy technique has clear safety benefits compared over craniotomy for surgical biopsy of brain tissue. Moreover, the finding that among the three acoustic pressures the lowest pressure (1.48 MPa) achieved the highest eGFP mRNA expression level is important. It suggests that vascular damage associated with FUS treatment at the relatively higher acoustic pressures (2.10 MPa and 3.34 MPa) may hinder the efficient passage of tumor biomarkers into the peripheral circulation. It also suggests that successful biomarker release may be achievable with acoustic pressures lower than 1.48 MPa, which can improve the safety of the FUS technique. Relatively higher acoustic pressures (1.45-3.34 MPa) were used in the current study than what is commonly used for the BBB disruption (e.g., 0.45 MPa), because in this study it was intended to enhance the interaction between microbubbles and the brain tissues. Optimization of the acoustic parameters (e.g., acoustic pressure, pulse repetition frequency, pulse length, and sonication duration) and microbubble parameters (e.g., size and dose) can be performed to explore the potential to achieve enhanced biomarker release with minimal or no tissue damage.

[0177] The disclosed technique can be optimized by assessing the efficiency of biomarker release under different FUS and microbubble parameters (see e.g., Example 2). Specifically, the release of biomarkers using FUS with lower pressures (e.g., 0.45 MPa) that are commonly used for BBB disruption without causing vascular damage can be evaluated. Second, the short-term and long-term safety of the FUS brain liquid biopsy technique can be assessed. Extracranial metastases of glioblastoma are rare^{sup.31}, and risk for tumor spread after surgical brain biopsies is considered

negligible. It is expected that the FUS liquid biopsies will not induce metastatic spread. Nevertheless, studies can be performed to determine the potential of metastasis associated with the FUS treatment. Third, the terminal cardiac puncture was used for blood collection as mice have a small total blood volume (~1.5 mL). Repeated blood sample collection was not feasible using the mouse model, but could be accomplished in humans and larger animals. Larger animal models (e.g., rats, pigs) can be used to collect blood samples at multiple time points after the FUS treatment. These blood samples can be used to assess the temporal dependency of the amount of biomarkers released by the FUS treatment and establish the optimal blood collection time. The larger animal models can be used to evaluate the repeatability of this technique by performing the FUS treatment on the same animal on multiple days. Fourth, the contrast-enhanced MR images acquired before and after the FUS treatment were quantified (see e.g., FIG. 1D). There was no significant difference in the MR contrast enhancement ratios among the three pressure groups (1.48 MPa, 2.10 MPa, and 3.34 MPa). This finding was not consistent with the quantification results of eGFR mRNA expression level shown in FIG. 3A-B, which found the eGFR mRNA expression level in the 1.48 MPa group was significantly higher than that of the other two higher pressure groups. Future studies will determine when lower acoustic pressures are used whether contrast-enhanced MRI is a useful tool in predicting the amount of biomarkers released by FUS. Fifth, the molecular marker (eGFP) was used to demonstrate the feasibility of FUS in noninvasive and spatially-focused liquid biopsy. Studies can use the same methods to assess the other tumor markers (e.g., DNA-based markers). The exact mechanism by which FUS-enabled release of molecular biomarkers is unknown. It is presently believed that the mechanism may be that the FUS-induced BBB disruption opens a “two-way door,” allowing “two-way trafficking” between the brain and blood. Future studies will explore the potential mechanisms of FUS-enabled biomarker release.

Conclusions

[0178] It was demonstrated that the combination of FUS and microbubbles allows detection of tumor-specific eGFP mRNA in the bloodstream that is otherwise undetectable. The presently disclosed findings established that FUS-mediated BBB disruption could enhance brain-to-blood trafficking. FUS may offer an enabling technique for noninvasive and regionally-specific brain tumor liquid biopsy that can be utilized in personalized brain cancer patient care.

Methods

Orthotopic Mouse Glioblastoma Models

[0179] All animal procedures were reviewed and approved by the Institutional Animal Care and Use Committee of Washington University in St. Louis in accordance with the National Institutes of Health Guidelines for animal research.

[0180] Two orthotopic mouse glioblastoma models were developed: (i) NCI athymic NCr-nu/nu mice (Strain 553, Charles River Laboratory, Wilmington, MA, USA) injected with U87 human glioblastoma cells; and (ii) NIH Swiss mice (Strain 550, Charles River Laboratory, Wilmington, MA, USA) implanted with GL261 murine glioblastoma cells. Mice were anesthetized and fixed into a stereotactic head frame. A paramedian incision was made on the scalp, and a 1-mm burr hole was drilled 2 mm posterior and 1.5 mm lateral to the bregma. eGFP-Luciferase-transduced glioblastoma cells (U87 or GL261) were mixed with Corning™ Matrigel (Catalog 356231, Corning Life Science, New York, USA) and injected through the burr hole using a syringe. The burr hole was sealed with bone wax, and the skin incision was glued together with tissue glue. The growth of the tumor was monitored using IVIS© Spectrum In Vivo Imaging System (Model 124262, PerkinElmer, Ohio, USA) once every week for four weeks. At around fifth week after tumor cell implantation, mice were recruited in the study described below.

Ultrasound Imaging-Guided FUS Treatment of U87 Mice

[0181] A total of nine mice with orthotopic U87 glioblastoma tumors were divided into two groups: treatment group (n=6) and control group (n=3). The treatment group was treated with FUS spatially

targeted at the tumor site after intravenous injection of microbubbles. The control group received no FUS. For FUS treatment, mice were first anesthetized with 2% isoflurane and placed on a stereotactic frame. A FUS system (VIFU 2000; Alpinion US Inc., Bothell, WA, USA) sonicated the tumor using the following parameters: frequency=1.5 MHz, peak negative pressure=3.28 MPa, pulse length=10 ms, pulse repetition frequency=1 Hz, duration=30 s at each location, and 4 separate locations were treated for each tumor (see e.g., FIG. 1A and FIG. 1E). The pressure amplitudes and beam dimensions of the FUS transducer were calibrated using a needle hydrophone (Onda, CA, USA) in a degassed water tank before the experiment. The pressures reported here were the peak negative pressures measured in water. The full width at half maximum (FWHM) of the axial beam and lateral beam were 6.04 mm and 0.62 mm, respectively. Before FUS sonication, microbubbles manufactured in house^{sup.32} were injected into the mouse tail vein at a concentration of 8×10^8 microbubbles per mL and a volume of 30 μ L per mouse. The in-house manufactured microbubbles comprised of a 90 mol % 1,2-distearoyl-sn-glycero-3-phosphocholine (DSPC) and 10 mol % 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(polyethylene glycol)2000](DSPE-PEG2000) (Avanti Polar Lipids, Alabaster, AL, USA) lipid-shell and a perfluorobutane (FluoroMed, Round Rock, TX, USA) gas-core. These microbubbles had a median diameter of 4-5 μ m, which were isolated from a poly-dispersed microbubble distribution using a differential centrifugation method^{sup.33}.

MRgFUS Treatment of GL261 Mice

[0182] A total of 12 mice with implantation of GL261 glioblastoma tumors in the brain were split into four groups: control group (n=3) and three treatment groups (n=3 for each group). The three treatment groups were treated using a clinical MRgFUS system (Sonalleve V2, Profound Medical Inc., Mississauga, Canada) equipped with a dedicated small animal adapter (FUS Instruments Inc., Toronto, Ontario, Canada) (see e.g., FIG. 1C)^{sup.34}. The Sonalleve MRgFUS system included a 256-element phased array transducer mounted to a five-axis robot positioner and located inside a modified MRI patient table. The acoustic fields generated by the phased array transducer were calibrated by a fiber optic hydrophone using the previously published method^{sup.35}. The FWHM of the axial beam and lateral beam were 12.10 mm and 1.37 mm, respectively. The MRgFUS system was integrated into a clinical MRI scanner (Ingenia 1.5T, Philips, Best, the Netherlands). The small animal adapter included a frame to hold an animal, a water reservoir, and a small animal MRI coil (imaging probe).

[0183] For the treatment of GL261 mice, mice were anesthetized with 1-2% isoflurane and placed on the small animal adapter. Optimark (gadoversetamide, a gadolinium-based contrast agent, 0.5 mmol/ml) was injected intravenously into the mice through the tail vein at a dose of 0.1 mmol/kg. Contrast-enhanced 3D, T₂-weighted MRI images of the mouse brain were acquired for treatment planning, and the FUS targeted location was selected to be the center of the tumor (see e.g., FIG. 1D). After intravenous injection of the same dose of microbubbles as that was used for U87 mice, the GL261 mice were treated by the MRgFUS system using the following parameters: frequency=1.44 MHz, pulse length=10 ms, pulse repetition frequency=1 Hz, and duration=2 min. The three treatment groups were treated at different peak negative pressures: 1.48 MPa, 2.10 MPa, and 3.34 MPa, respectively. After treatment, contrast-enhanced MRI images were acquired for confirming successful tumor targeting by the FUS (see e.g., FIG. 1D). Blood samples of 500-800 μ L were collected from the heart about 20 min after the FUS treatment and prepared for qPCR analysis of eGFP mRNA.

Analysis of eGFP mRNA

[0184] Blood samples of 500-800 μ L were collected from the heart about 4 min (U87) or 20 min (GL261) after the FUS treatment and prepared for qPCR analysis of eGFP mRNA. All the collected blood was centrifuged at 3,000 rpm for 10 minutes. The supernatant was collected. RNA samples were purified using the miRNeasy serum/plasma kit (Catalog no. 217184, Qiagen, USA).

Agencourt RNAClean XP beads (Catalog no. A63987, Beckman Coulter Inc., USA) were used to

further purify the RNA. The RNA was then reverse transcribed to cDNA using the Applied Biosystems™ high-capacity cDNA reverse transcription kit (Catalog no. 4368814, Thermo Fisher Scientific, USA). Two eGFP primer pairs (TABLE 1) were designed using OligoPerfect™ Designer (ThermoFisher Scientific, USA). 5.8s rRNA was used as an internal amplification control with its forward and reverse primer sequences also listed in TABLE 1. The quantitative real-time PCR was performed using SYBR™ Green PCR master mix (Applied Biosystems™). All of the PCRs were performed on a 7900HT Fast Real-Time PCR System (Catalog #4329001, Thermo Fisher Scientific, USA) using the following protocol: the reaction mixture was heated at 95° C. for 10 min, followed by 40 cycles of 95° C. for 5 s and 60° C. for 30 s. Amplification and dissociation curves generated by the SDS2.3 (Applied Biosystems) software were used for gene expression analysis.

[0185] Duplicate reactions were run for each sample and each primer set. 5.8S rRNA was used as the internal control to normalize the PCR data by calculating cycle threshold change ($\Delta C_{sub.T}$) by subtracting $C_{sub.T}$ of the eGFP ($C_{sub.T,eGFP}$) by the $C_{sub.T}$ of housekeeping gene, 5.8s rRNA ($C_{sub.T,5.8S}$). The gene expression level was determined using the $2^{-\Delta CT}$ method: $2^{-\Delta C_{sub.T}} = 2^{-(C_{sub.T,eGFP} - C_{sub.T,5.8S})}$. Maximum cycle number was set to 40. In order to assess the reliability of 5.8S rRNA as the internal control, $2^{-C_{sub.T}}$ was calculated and compared for the treated and control groups. The $2^{-C_{sub.T}}$ of the control mice ($1.8 \times 10^{-3} \pm 8.6 \times 10^{-4}$) was not significantly different from the $2^{-C_{sub.T}}$ of the FUS-treated mice ($1.1 \times 10^{-3} \pm 1.8 \times 10^{-3}$; $p=0.38$; two-tailed Mann-Whitney U test).

Histological Analysis

[0186] After blood collection, all the mice were transcardially perfused with 0.01 M phosphate-buffered saline and then with 4% paraformaldehyde, and their brains were harvested and prepared for paraffin sectioning. The mouse brains were horizontally sectioned to 15 μ m slices and used for H&E staining.

Statistical Analysis

[0187] GraphPad Prism (v6.04, La Jolla, CA, USA) and R (v3.4.1, R Core Development Team, 2017) were used to perform statistical analyses. Given the non-normality of $2^{-\Delta CT}$ distribution (Shapiro-Wilk test <0.05), group comparison was made using a non-parametric Mann Whitney U Test. A p -value <0.05 was used to determine statistical significance.

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Example 2: Optimization of Focused Ultrasound-Enabled Brain Tumor Liquid Biopsy (FUS-LBx) [0225] The following example describes the optimization of the FUS-LBx technique by investigating the effects of FUS acoustic pressure on the tumor biomarker release level and potentially associated hemorrhage burden.

[0226] The development of noninvasive approaches for brain tumor diagnosis and monitoring continues to be a major medical challenge. Although blood-based liquid biopsy has received considerable attention in various cancers, limited progress has been made for brain tumors, at least partly due to the hindrance of tumor biomarker release into the peripheral circulation by the blood-brain barrier. As shown in Example 1, the use of focused ultrasound-enabled brain tumor liquid biopsy (FUS-LBx) was demonstrated. The objective of this example was to optimize the FUS-LBx technique by investigating the effects of FUS acoustic pressure on the tumor biomarker release level and potentially associated hemorrhage burden. Mice with orthotopic implantation of enhanced green fluorescent protein (eGFP)-transfected murine glioma cells were treated using magnetic resonance (MR)-guided FUS system in the presence of systemically-injected microbubbles at three peak negative pressure levels (0.59, 1.29, and 1.58 MPa). Plasma eGFP mRNA levels were quantified with the quantitative polymerase chain reaction (qPCR). Contrast-enhanced MR images were acquired before and after the FUS treatment. FUS at 0.59 MPa resulted in increased plasma eGFP mRNA levels, comparable to those of higher acoustic pressures (1.29 MPa and 1.58 MPa). Hemorrhage associated with FUS at 0.59 MPa was significantly lower than that with higher acoustic pressures and not significantly different from the control group. MRI analysis revealed that post-sonication intratumoral and peritumoral hyperenhancement were associated with the level of FUS-induced biomarker release and the extent of hemorrhage. Taken together, this study suggests that FUS-LBx technique can be optimized to be a safe and effective image-guided biomarker release technique.

Introduction

[0227] Central nervous system (CNS) tumors are significant causes of cancer morbidity and mortality, especially in children and young adults where they account for ~20-30% of cancer deaths.^{sup.1} Noninvasive neuroimaging modalities (e.g., magnetic resonance imaging (MRI), and computerized tomography (CT)) are used to evaluate tumor lesions, but observed changes, especially after treatment, can be difficult to interpret.^{sup.2} Surgical resection or stereotactic biopsy is typically performed for histologic confirmation and increasingly for genetic profiling. However, tissue biopsy requires brain surgery and can be associated with adverse effects such as hemorrhage and infection.^{sup.3} Repeated tumor biopsies that may be needed for tracking tumor evolution, treatment response, and tumor recurrence are often not feasible. Furthermore, tissue biopsies may be challenging when tumors are located at difficult locations in the brain or CNS or patients that are too ill to tolerate invasive procedures.^{sup.4}

[0228] Noninvasive blood-based liquid biopsies are a rapidly emerging strategy to provide genetic tumor profiling that can be used for treatment selection, treatment monitoring, residual disease detection, and asymptomatic individual screening.^{sup.5} Some success has been achieved in integrating blood-based liquid biopsy into the routine clinical diagnostics.^{sup.6,7} However, limited progress has been made for brain tumors. Although several groups have reported biomarker detection in the cerebrospinal fluid (CSF) of patients with brain tumors.^{sup.8,9} obtaining CSF via lumbar puncture remains an invasive procedure and is often unsafe in these settings. Blood-based liquid biopsies are noninvasive, but there remain multiple hurdles for their application in brain tumors. First, the blood-brain barrier (BBB) restricts the release of large molecules such as DNA and RNA from the tumor to the peripheral circulation.^{sup.10} Even though partial BBB disruption is a core feature of glioblastoma (the most common type of high-grade glioma) at the late stage of the tumors, the BBB remains intact in the early stage and the low-grade diffuse gliomas.^{sup.11} Second, brain tumors, especially glioblastomas, are heterogeneous with spatially distinct molecular

profiling.^{sup.12} Localization of tumor areas that harbor specific mutations is not possible using conventional methods of liquid biopsy, as these methods are inherently spatially agnostic. Third, many tumor markers, such as some cell-free RNAs.^{sup.13} and DNAs.^{sup.14}, have short half-lives in blood. Detection of these biomarkers can be enhanced by stimulating their release from the tumor to the circulation and precisely controlling the blood-collection time to be shorter than their lifetimes in the blood. Therefore, a noninvasive, spatially resolved, and temporally controlled liquid biopsy technique is in great need to improve the clinical care of patients with CNS tumors.

[0229] Focused ultrasound-enabled brain tumor liquid biopsy (FUS-LBx) (see e.g., Example 1) offers a technique that can provide noninvasive, spatially resolved, and temporally controlled brain tumor liquid biopsies. FUS has emerged as a technology with the potential to noninvasively exert mechanical and thermal effects on the brain tissue. When coupled with intravenously injected microbubbles, low-intensity pulsed FUS-induced mechanical effects can transiently and non-invasively open the BBB without causing any vascular or tissue damage.^{sup.16-18} In Example 1, it was demonstrated that FUS in combination with microbubbles could also be exploited to release biomarkers from brain tumors in murine glioblastoma models.^{sup.15} This approach is different from previously reported FUS-facilitated liquid biopsy techniques developed for in vivo biomarker release from cancers outside the brain.^{sup.19-21} In one of those reports, pulsed high-intensity focused ultrasound (HIFU) with high acoustic pressures (ultrasound frequency=1.5 MHz, peak compressional focal pressure=90 MPa, and peak rarefactional focal pressure=17 MPa) was used to induce histotripsy (i.e., a technique for mechanical tissue fractionation) in a rat model of prostate cancer, and this enhanced the release of cell-free tumor microRNA into the blood circulation.^{sup.19} In another study, a chicken embryo tumor model was used to show the feasibility of amplifying the release of extracellular vesicles using HIFU (ultrasound frequency=1.15 MHz and peak to peak pressure was within the range of 10-30 MPa) in combination with phase-change nanodroplets which changed to microbubbles upon HIFU sonication.^{sup.20} In a recent study, two protein biomarkers were found to be significantly increased in the plasma of patients undergoing HIFU thermal ablation (ultrasound frequency=1.1 MHz and power of 100-200 W) of uterine fibroids.^{sup.21} All these previous studies used HIFU to induce permanent mechanical or thermal disruption of the tumor to enhance the release of tumor biomarkers. The tissue damaging effect limits the application of these techniques as diagnostic tools in a sensitive organ such as the brain. In contrast, the FUS-LBx technique described here uses low-intensity pulsed FUS, which has the potential advantage of enabling the biomarker release without causing tissue damage. However, in Example 1, the acoustic pressures used were intentionally selected to be relatively high (1.52-3.53 MPa) to increase the chance of success in releasing biomarkers. As expected, hemorrhage was found in the FUS-targeted brain area.

[0230] The objective of this example was to investigate the effects of FUS acoustic pressure on the level of tumor biomarker release and the extent of associated hemorrhage in order to optimize the FUS-LBx technique. It was sought to determine the optimal ultrasonic pressure for FUS-LBx that can sufficiently increase plasma levels of the tumor biomarkers while at the same time minimize the risk of hemorrhage in the brain. It was further explored whether post-sonication changes in tumor MR contrast enhancement can predict successful biomarker release for the future development of image-guided FUS-LBx technique.

Results

Effect of the Peak Negative Pressure on FUS-LBx Yield

[0231] Mice with orthotopic implantation of enhanced green fluorescent protein (eGFP)-transfected murine glioblastoma cells were recruited into this study when the maximum diameter of the tumor reached 2 mm. The mice were treated by an MR-guided FUS system (see e.g., FIG. 5A, FIG. 5B) at three different peak negative acoustic pressure (PNP) levels: (i) 0.59 MPa; (ii) 1.29 MPa; (iii) 1.58 MPa (n=5 in each group). The acoustic pressure output of the FUS transducer was calibrated using a fiber-optic hydrophone following a previously published method (see e.g., FIG. 5C).^{sup.22}

[0232] Two primer sets were used to improve the robustness of the quantitative polymerase chain reaction (qPCR) results. When compared to the control group (no FUS or microbubble), all three FUS-treated groups demonstrated significant increases in plasma eGFP mRNA levels (all FUS-treated mice vs. control group: one-tailed $p=6.5 \times 10^{-5}$ for both primer sets: primers A and B; TABLE 2, FIG. 6A-B). When compared with the control group, there were 55-fold (Primer A: one-tailed $p=0.004$) and 221-fold (Primer B: one-tailed $p=0.004$) increase in plasma eGFP mRNA levels in mice that received FUS with the lowest PNP (0.59 MPa). Plasma eGFP mRNA levels in mice treated with 1.29 MPa and 1.58 MPa achieved respectively about 2,000-fold and 8,000-fold average enhancement relative to the control group, respectively (TABLE 4). For primer A, plasma eGFP mRNA levels were significantly greater with high acoustic pressures when compared to 0.59 MPa-FUS treated mice (1.29 MPa vs. 0.59 MPa: 35-fold higher, one-tailed $p=0.004$; 1.58 MPa vs. 0.59 MPa: 151-fold higher, one-tailed $p=0.004$). These effects were less pronounced when eGFP mRNA levels were measured using primer B (1.29 MPa vs. 0.59 MPa: 5-fold higher, one-tailed $p=0.048$; 1.58 MPa vs. 0.59 MPa: 22-fold higher, one-tailed $p=0.11$).

TABLE-US-00004 TABLE 2 Group average $\pm$ standard deviation of eGFP RNA levels, hemorrhage densities, and tumor MRI enhancement.

Group	Control	0.59 MPa	1.29 MPa	1.58 MPa
eGFP mRNA level	$2.2 \times 10^{-8} \pm 3.1 \times 10^{-8}$	$1.2 \times 10^{-6} \pm 9.1 \times 10^{-7}$	$4.2 \times 10^{-5} \pm 4.7 \times 10^{-5}$	$1.8 \times 10^{-4} \pm 9.1 \times 10^{-7}$
(Primer A), $2^{-\Delta CT}$ eGFP mRNA level	$2.2 \times 10^{-8} \pm 1.5 \times 10^{-8}$	$4.8 \times 10^{-6} \pm 5.5 \times 10^{-6}$	$2.5 \times 10^{-5} \pm 2.7 \times 10^{-5}$	$1.0 \times 10^{-4} \pm 1.7 \times 10^{-4}$
(Primer B), $2^{-\Delta CT}$	Hemorrhage	0.12 ± 0.06	0.17 ± 0.08	0.95 ± 0.6
density (%)	Tumor MR	12.5 ± 4.1	13.7 ± 3.9	13.5 ± 3.6
enhancement		13.7 ± 2.2		

Effect of the Peak Negative Pressure on Brain Hemorrhage

[0233] Hematoxylin and eosin (H&E) stained sections of the brain were examined for the presence of microhemorrhages. Representative images of the H&E-stained sections are shown in FIG. 7A-FIG. 7D. After color deconvolution, areas of microhemorrhage were identified using the positive-pixel count algorithm (see e.g., FIG. 7E). Microhemorrhage density was calculated as the proportion of surface area of microhemorrhage to the total surface area of the evaluated brain tissue in a given slice. There was no significant difference in microhemorrhage density between mice treated with 0.59 MPa FUS and the control mice (TABLE 4, FIG. 7F). The microhemorrhage density was significantly higher in 1.29 MPa and 1.58 MPa groups than the 0.59 MPa and control groups. Microhemorrhages seen in the control mice were predominantly scattered in the tumor. In addition to scattered intratumoral microhemorrhage, peritumoral hemorrhage near the interface of the tumor and normal brain parenchyma was seen in 1 out of 5 mice treated with 0.59 MPa FUS as well as in 4 out of 5 mice in the 1.29 MPa and 1.58 MPa FUS-treated groups. Large variations in the hemorrhage density were observed within each group, especially at higher pressures (1.29 MPa and 1.58 MPa).

Predicting FUS-LBx Yield Using MRI

[0234] Increased MRI image intensity on contrast-enhanced MR images was seen both within the tumor and in the peritumoral area of the tumor after sonication (see e.g., FIG. 8A). Because there is a significant difference between intratumoral and peritumoral enhancement before sonication, it was opted to evaluate post-sonication intratumoral and peritumoral enhancement separately. Possible associations between post-sonication enhancements and microhemorrhage density and FUS-LBx yield were interrogated. Greater sonication-induced intratumoral and peritumoral contrast enhancement were associated with higher hemorrhage burden (see e.g., FIG. 8B, FIG. 8C) and greater post-sonication plasma eGFP mRNA levels detected using primer A (see e.g., FIG. 8D, FIG. 8E). The correlation coefficients were lower for the eGFP levels ($r=0.52$ for intratumoral contrast enhancement and $r=0.64$ for peritumoral enhancement) than the hemorrhage burden ($r=0.71$ for intratumoral contrast enhancement and $r=0.74$ for peritumoral enhancement). Similar associations were seen between intratumoral/peritumoral enhancements and eGFP mRNA levels

measured with primer B ($r=0.51$ for intratumoral contrast enhancement and $r=0.56$ for peritumoral enhancement). The correlation coefficients for all the above analysis were slightly higher with peritumoral contrast enhancement than intratumoral contrast enhancement.

DISCUSSION

[0235] In a murine orthotopic glioblastoma model, it was demonstrated that FUS-LBx technique could be applied with low acoustic pressure without significant sonication-induced hemorrhage. The findings from the prior study^{sup.15} (Example 1) were replicated and expanded and used PNPs (0.59-1.58 MPa) that were lower than those in the previous study (1.52-3.53 MPa). Within 0.59-1.58 MPa, higher pressures were associated with a trend of higher biomarker release level but also increased risk of brain hemorrhage.

[0236] The results of this study demonstrated that the FUS-induced release of eGFP mRNA from the tumor into the blood circulation is higher when FUS with high PNPs (1.29 and 1.58 MPa) is applied compared with relatively low pressure (0.59 MPa) (see e.g., FIG. 6A-B). The reason for this difference might be that higher PNPs may lead to an increased likelihood of microbubbles undergoing violent inertial cavitation that can, in turn, induce greater disruption of the BBB.^{sup.23-26} In this way, there is a higher possibility that a higher amount of eGFP mRNA can be released to the bloodstream. In Example 1, the eGFP mRNA level was found to not be pressure dependent within the range of 1.52-3.53 MPa. Taken together, these findings suggest that as FUS pressure increases the biomarker release level increases and then reaches a plateau. This may be in part due to potential paradoxical effects of increasing PNP on FUS-LBx yield. On the one hand, higher FUS PNP can result in greater BBB disruption, while on the other hand, it can result in vascular damage and decreased tumor perfusion, and hinder tumor biomarker release into peripheral circulation. Moreover, the relative expression levels of eGFP (normalized to 5.8S rRNA) were used in the plasma to index FUS-LBx diagnostic yield. It is possible that at higher PNPs, potentially increased levels of eGFP are offset by possible increases in 5.8S rRNA released from damaged brain tissue.

[0237] For the evaluation of the brain hemorrhage, the results showed a significantly higher microhemorrhage when comparing the 1.29 MPa and 1.58 MPa group with the 0.59 MPa group or the control group (see e.g., FIG. 7F). Although there was no statistically significant difference in the extent of microhemorrhage between the control group and the FUS-treated group receiving 0.59 MPa PNP, microhemorrhage beyond the bounds of the tumor was observed along the interface between the tumor and normal brain tissue in one out of five cases. This pattern of hemorrhage was more pronounced with higher FUS PNPs (see e.g., FIG. 7C and FIG. 7D). This pattern of hemorrhage in the tumor periphery may result from the immaturity and instability of peritumoral vasculature rendering them more susceptible to mechanical injury induced by FUS.

[0238] Relatively large variations within each group were observed in the analysis of both FUS-LBx yield and FUS-induced hemorrhage. Such variations among repeated FUS treatments have been reported before in FUS-induced BBB disruption experiments. It can be caused by variations in parameters that are hard to control, such as the size distribution of microbubbles that reach the targeted brain location.^{sup.27}, circulating microbubble concentration in blood.^{sup.28}, blood vessel density within the treated region of the tumor.^{sup.29}, and heterogeneous acoustic property of skull.^{sup.30} Therefore treatment monitoring techniques, such as passive cavitation imaging.^{sup.31-33}, can be used to detect and characterize potential variations among repeated FUS treatment. Feedback control algorithms can also be implemented based on cavitation monitoring to control the FUS output during the treatment to minimize variations associated with the FUS treatment.^{sup.34,35}

[0239] This study also demonstrated that greater contrast enhancement was associated with both a greater level of eGFP plasma level and a higher hemorrhage density (see e.g., FIGS. 8A-E). Post-sonication intratumoral and peritumoral enhancements were both associated with higher released tumor markers in plasma. At the same time, post-sonication enhancement within the tumor and in its periphery were associated with microhemorrhage density. MR imaging markers that can predict

safe and successful liquid biopsy are essential for translation of FUS-LBx to the clinic, as they can inform the clinician for sufficiency of FUS application for liquid biopsy or need for a higher magnitude of PNP. Future studies will be performed to explore the potential of MRI in the assessment of FUS-LBx outcome and safety. MRI can be combined with passive cavitation imaging to develop image-guided FUS-LBx which employs real-time passive cavitation imaging for FUS-LBx treatment monitoring and feedback control and utilizes MRI for treatment outcome and safety assessment.

[0240] There are limitations to be considered. First, plasma mRNA levels of eGFP were used that were uniquely expressed in the tumor cells to evaluate FUS-mediated liquid biopsy. While this approach is adequate to determine the effectiveness of FUS-LBx, it may not fully recapitulate the clinical settings for liquid biopsy. However, the efficacy of FUS-LBx can be interrogated to discern disease-causing variations of naturally occurring tumor-specific biomarkers.^{sup.36} Of note, biomarkers can be evaluated within a short interval from their release to the circulation after FUS application. This permits the detection of short-lived molecules such as mRNAs and some proteins. Second, similar to Example 1, blood was collected using terminal cardiac puncture as mice have a small total blood volume (~1.5 mL). This approach precludes repeated blood sampling to monitor plasma mRNA levels over time following sonication or to determine the effects of serial FUS-mediated BBB opening on plasma tumor biomarkers. Therefore, in order to determine the temporal pattern of released tumor biomarkers and feasibility of serial liquid biopsy, FUS-LBx needs to be evaluated in larger animal models such as rats, pigs, and non-human primates. Third, a few inconsistencies were observed in plasma eGFP mRNA findings with different primer sets. Although eGFP mRNA measured with both primer sets was overall highly correlated ($r=0.93$, root mean square difference=1.85 cycle threshold), the difference between plasma eGFP mRNA levels of mice treated 0.59 MPa and higher PNPs were less prominent with primer set B. More robust and reproducible measurements of RNA levels with droplet digital PCR.^{sup.37} could mitigate these issues and enable reliable detection of small differences in FUS-LBx.

Conclusions

[0241] This study showed that an acoustic pressure as low as 0.59 MPa was sufficient for FUS-LBx. Although higher peak negative acoustic pressures tended to be associated with a better yield of liquid biopsy, it was also associated with a higher burden of microhemorrhage. MRI can be used in not only guiding the FUS targeting of a specific brain region but also providing predictions of the biomarker release level and potential hemorrhage extent. Future studies are warranted to develop MRI-guided FUS-LBx to improve its safety and efficacy.

Methods

FUS-LBx Treatment Procedure

[0242] All animal studies were reviewed and approved by the Institutional Animal Care and Use Committee of Washington University in St. Louis in accordance with the National Institutes of Health Guidelines for animal research.

[0243] Mice (Strain 550, 6-8 weeks, $n=20$, Charles River Laboratory, Wilmington, MA, USA) were implanted with GL261 murine glioblastoma cells on day 0 using an established protocol.^{sup.15} The growth of the tumor was monitored using a clinical MRI scanner (Ingenia 1.5T, Philips Healthcare, Best, the Netherlands) coupled with a small animal coil (FUS Instruments Inc., Toronto, Ontario, Canada) as shown in FIG. 5A-B twice per week. Once the maximum diameter of the tumor reached 2 mm as measured based on the contrast-enhanced MRI, the mice were randomly assigned into four groups: control group that received no treatment ($n=5$) and three groups that received FUS treatment with three different PNP: 0.59 MPa ($n=5$), 1.29 MPa ($n=5$), and 1.58 MPa ($n=5$). All other parameters were the same across the treatment groups: 1.44 MHz, sonication duration: 240 s; treatment target: 1 (at the center of the tumor); pulse repetition frequency: 1 Hz; duty cycle: 1%; pulse length: 10 ms. All FUS sonications were performed on post-implantation days 15 or 16.

[0244] The details of the sonication procedure and the dose and concentration of the microbubbles have been described previously.^{sup.15} Briefly, the mice were treated by FUS using a clinical MR-guided FUS system (Sonalleve V2, Profound Medical Inc., Mississauga, Canada) that integrated a clinical MRI scanner (Ingenia 1.5T, Philips Healthcare, Best, the Netherlands) with a 256-element phased array FUS transducer. The system was configured for small animal study by coupling with a small animal adaptor (FUS Instruments Inc., Toronto, Ontario, Canada) placed above the acoustic window of the FUS transducer (see e.g., FIG. 5A and FIG. 5B). The adaptor consisted of a small animal coil, a standoff, and a mouse bed. The software of the MR-guided FUS system was modified to achieve tumor targeting under the guidance of MR images of mouse brains acquired using the small animal coil. It was also modified to perform pulsed FUS sonication at low-pressure levels used in this study by only using 128 elements. The acoustic fields generated by the phased array transducer was calibrated by a fiber optic hydrophone using previously published methods (see e.g., FIG. 5C).^{sup.22,38} The full width at half maximum of the axial beam and lateral beam was 12.10 mm and 1.37 mm, respectively.

[0245] Contrast-enhanced T1-weighted turbo spin-echo MR images (TR, 500 ms; TE, 13 ms; acquisition matrix, 96×96; resolution, 0.2 mm×0.2 mm×0.5 mm) were acquired before and after the FUS treatment to quantify changes in intratumoral and peritumoral enhancement.

Plasma eGFP mRNA Level Quantification

[0246] Blood samples of 500-800 µL were collected from the heart about 20 min after FUS sonication and prepared for qPCR analysis of eGFP mRNA. The methods of qPCR analysis of eGFP mRNA have been described in Example 1. Briefly, RNA was extracted from the plasma samples using miRNeasy serum/plasma kit (Catalog no. 217184, Qiagen, USA) followed by Agencourt RNAClean XP beads (Catalog no. A63987, Beckman Coulter Inc., USA). Extracted RNA was then converted to cDNA using the Applied Biosystems high-capacity cDNA reverse transcription kit (Catalog no. 4368814, Thermo Fisher Scientific, USA). Two primer sets were used to quantify eGFP levels. 5.8S rRNA was used as the internal control to normalize the PCR data by calculating cycle threshold change ($\Delta C_{sub.T}$) by subtracting $C_{sub.T}$ of the eGFP ($C_{sub.T,eGFP}$) by the $C_{sub.T}$ of the housekeeping gene, 5.8s rRNA ($C_{sub.T,5.8S}$). The relative gene expression level was determined using the $2^{-\Delta C_{sub.T}}$ method: $2^{-\Delta C_{sub.T}} = 2^{-(C_{sub.T,eGFP} - C_{sub.T,5.8S})}$.

MRI Image Analysis

[0247] MR image processing and analysis were performed using tools available in FSL v5.0.10.^{sup.36} First, tumor regions were segmented semi-automatically on the pre-sonication contrast-enhanced T1-weighted images using ITK-SNAP v3.6.0.^{sup.35} Second, on each pre-sonication contrast-enhanced T1-weighted MRI image of the brain tumor, a spherical control mask with 1 mm radius was drawn in the normal appearing brain and its mean and standard deviation were calculated. Third, all contrast-enhanced T1-weighted images (both pre-sonication and post-sonication images) were intensity normalized by subtracting the mean and then dividing by the standard deviation of the signal intensity within the control mask. Fourth, the peritumoral area was defined as the brain regions within 2 mm vicinity of the tumor. A preliminary mask was created using the `distancemap` command in FSL. This mask was then edited manually to exclude voxels outside the brain parenchyma. Finally, to quantify post-sonication changes in MRI contrast enhancement, the difference between normalized post-sonication and pre-sonication T1-weighted images was calculated within the intratumoral and peritumoral area masks.

Histologic Analysis

[0248] After blood collection, all mice were transcardially perfused with 0.01 M phosphate-buffered saline (PBS) followed by 4% paraformaldehyde. Brains were harvested and prepared for paraffin sectioning. The mouse brains were horizontally sectioned to 15 µm slices and used for H&E staining. Whole slide images of H&E tissue sections were digitally acquired using Nanozoomer 2.0-HT slide scanner (Hamamatsu Photonics, Hamamatsu City, Japan). For each

mouse, the H&E stained slice with the largest tumor surface area and least artifact was selected for quantitative analysis. QuPath v0.1.3.sup.39 was used to detect areas of microhemorrhage. After color deconvolution (hematoxylin vs. eosin), areas of microhemorrhage was detected using the positive-pixel count algorithm. Microhemorrhage density was calculated as the percentage of microhemorrhage surface area over the entire evaluated surface area (%).

Statistical Analysis

[0249] All statistical analyses were performed using R statistical software v3.5.0 (<https://www.R-project.org/>).sup.40 and graphically displayed using ggplot2 package v2.2.1.sup.41. For group comparisons, the non-parametric Mann-Whitney U test was used. Associations between two continuous variables were assessed using Pearson correlation analysis. eGFP plasma levels were log transformed for correlation analysis. Statistical significance was set at $p < 0.05$.

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Example 3: Noninvasive and Localized Blood-Brain Barrier Opening and Detection of Brain-Specific Biomarkers

[0291] This example shows the pressure field measurements of the designed transducer and the evaluation of its performance and the detection of brain-specific biomarkers.

[0292] Blood-brain barrier (BBB) opening with focused ultrasound and microbubbles (MB) have been demonstrated to be a noninvasive, targeted, safe, and effective technique mostly in anesthetized animal models. However, the requirement of anesthesia can limit the future clinical application of this noninvasive technology for the treatment of brain diseases as patient real-time feedback is critical to ensure the safety of the treatment. So far, only one study demonstrated the feasibility of BBB opening using FUS in awake non-human primates after training the non-human primates to cooperate during the treatment. No study has evaluated the feasibility of FUS-induced

BBB opening in the most commonly used animal model—mice, because of the technical challenges associated with designing a device for FUS sonication in awake mice.

[0293] The blood-brain barrier (BBB) blocks large molecules (>400 Da) from entering the central nervous system (CNS). Focused Ultrasound (FUS) combined with microbubble is the only available technique to non-invasively, locally, and transiently open the BBB. Previous preclinical research using small animal models has demonstrated this technique as a promising way to treat various brain disorders. This technique has also been tested on non-human primates with success. Clinical Studies with FUS are actively investigated for treatment of brain tumor and Alzheimer's disease. Besides its clinical translation potential, this technique has a broad application in preclinical research. It can be used to delivery various agents to the brain, such as chemo drugs, proteins, peptides, nanoparticles, gene vectors, and stem cells. However, currently there is a lack of devices that are affordable, high throughput, and dedicated for small animal research. Here is disclosed a new FUS-induced BBB opening device that meets these needs. The transducer was made using an inexpensive flat ultrasound transducer coupled to a 3D printed acoustic lens. The cost of the transducer was about \$80, while the FUS transducer commonly used in preclinical research can cost over \$10,000. Multiple transducers can be used simultaneously to achieve high throughput treatment of multiple mice at the same time.

[0294] A wearable helmet was designed for non-invasive, targeted FUS sonication of the mouse brain while the animals were awake. The helmet had a modular design featuring easy removal and installing of the unit for targeting different brain regions. The performance of the helmet in inducing BBB opening at the caudate putamen was assessed in four awake mice and four anesthetized mice in the presence of intravenously injected microbubbles. Evan's blue was co-injected with the microbubbles for the evaluation of the BBB permeability using fluorescence imaging of ex vivo brain slices.

[0295] The whole helmet with ultrasound transducer and wirings weighed 6.6 g. The constraint design of the helmet minimized the effect of mouse movement on targeting. The helmet achieved localized BBB opening at the targeted brain location. The fluorescence intensity of the Evan's blue in the brains of awake mice was higher than that in anesthetized mice, suggesting that FUS-induced BBB opening was affected by anesthesia.

[0296] The helmet design of the FUS device provides an innovative tool to study FUS-induced BBB opening in awake mice.

[0297] FIG. 9 is a photo of the transducer. FIG. 10 is a photo of a mouse treated using the device. FIG. 11 is a photo showing the successful drug delivery to the mouse brain, indicated by the leakage of the blue color. FIG. 12 is an image of spatially precise drug delivery to the hippocampus, indicated by the enhanced fluorescence observed on the right side of the of the mouse hippocampus.

[0298] An MRgFUS system was also designed for a pig (see e.g., FIG. 13). This design can be easily configured for clinical use by humans. The MRgFUS system is designed to integrate into the bore of an MRI scanner (see e.g., FIG. 13).

[0299] The system allows for planning and targeting of image-guided focused ultrasound liquid biopsy (FUS-LBx) (see e.g., FIG. 14); monitoring and feedback control using an US sensor to monitor microbubble response to US (see e.g., FIG. 15); and outcome assessment (see e.g., FIG. 16).

[0300] Described herein is also a method of monitoring cavitation (e.g., microbubble behavior) (see e.g., FIG. 15). The FUS system comprised a sensor that measures acoustic signal. This monitoring informs if the pressure should be increased or decreased by measuring the acoustic signal. Here, the IC line (blue) monitors the bubbles and the SC shows the active calibration to achieve the amount of pressure delivery. This can be used to monitor for safety.

[0301] Another routine MRI is performed to detect the leakiness of the area that was administered US.

[0302] FUS-LBx was performed in a large animal model (pig). FIG. 13 shows an image of the pig MRgFUS system. Brain-specific biomarkers, glial fibrillary acidic protein (GFAP), myelin basic protein (MBP), and S100 calcium-binding protein B (S100B) were measured before and after FUS. Increased levels of the brain specific biomarkers were shown in increased levels post-FUS (see e.g., FIG. 17).

Example 4: Sonobiopsy Release of Brain-Derived Protein Biomarkers from Spatially Targeted Brain Regions

[0303] The following example describes sonobiopsy releases brain-derived protein biomarkers from spatially targeted brain regions into the blood circulation in a tauopathy mouse model.

[0304] Background: Alzheimer's disease and other tauopathies are a heterogeneous class of neurodegenerative disorders characterized by the deposition of various pathogenic forms of tau protein in the brain. Much like cancer, identification of the molecular drivers and pathways of neurodegeneration is critical to individualized targeted treatment of these disorders. However, despite significant advances in fluid biomarker detection, characterization of these molecular subtypes is limited by the blood-brain barrier (BBB). Sonobiopsy employs MRI-guided focused ultrasound (FUS) with microbubbles to locally disrupt the BBB and release molecular biomarkers into the bloodstream for the molecular diagnosis of brain diseases by blood tests.

[0305] Purpose: To evaluate the feasibility and safety of sonobiopsy in the diagnosis of brain-derived protein biomarkers in a transgenic mouse model of tauopathy (PS19).

[0306] Materials and Methods: Sonobiopsy was performed by sonicating the cerebral hemisphere in 2-month-old PS19 and wild-type mice, followed by measurement of plasma phosphorylated tau (pTau) species. Next, spatially-targeted sonobiopsy was performed by sonicating either the cerebral cortex or the hippocampus in 6-month-old PS19 mice. To detect changes in plasma neurofilament light chain (NfL, a biomarker of neurodegeneration) levels, blood was collected before and after sonication. Histological staining was performed to evaluate tissue damage following sonobiopsy.

[0307] Results: In the 2-month-old mice, there were significant increases in normalized levels of plasma pTau species in the FUS-treated PS19 mice compared to the control PS19 mice (pTau-181/mouse-Tau [mTau] ratio: 1.7-fold-increase, $p=0.0058$; pTau-231/mTau ratio: 1.4-fold-increase, $p=0.048$). In the 6-month-old PS19 mice, spatially targeted sonobiopsy resulted in a 2.3-fold increase in plasma NfL following the sonication of the hippocampus and cerebral cortex ($p=2.2 \times 10^{-4}$). Optimization of the sonobiopsy parameters reduced the observed microhemorrhage density from $1.38 \pm 0.92\%$ to $0.52 \pm 0.32\%$.

[0308] Conclusion: This study demonstrated the feasibility of sonobiopsy to release pTau species and a key marker of neurodegeneration to the circulation for the diagnosis of neurodegenerative disorders.

Key Results

[0309] In a mouse model of tauopathy, focused ultrasound-mediated blood-based liquid biopsy (sonobiopsy) significantly changed the composition of circulating phosphorylated tau species in the bloodstream by increasing the normalized pTau-181 levels (1.7-fold-increase, $p=0.0058$) and, to a lesser extent, the normalized pTau-231 levels (1.4-fold-increase, $p=0.048$) compared with the control mice.

[0310] Spatially targeted sonobiopsy significantly increased the levels of neurofilament light chain proteins in the blood circulation by 2.3-fold ($p=2.2 \times 10^{-4}$) compared with the levels using conventional blood-based liquid biopsy.

[0311] Sonobiopsy was performed without evidence of acute brain tissue damage through parameter optimization.

Summary Statement:

[0312] Focused-ultrasound mediated blood-based liquid biopsy (sonobiopsy) enhances the release of biomarkers of neurodegeneration from the spatially-targeted brain regions into the blood circulation.

Introduction

[0313] A hallmark of age-related neurodegenerative disorders is the accumulation of abnormal protein species in the brain (1). In tauopathies, hyperphosphorylation of the tau protein leads to its dissociation from the microtubule, misfolding, aggregation, which ultimately triggers a secondary cascade of neurotoxic effects that culminates in neuronal death. Akin to driver mutations in brain tumors, there are considerable variations in the post-translational modifications of pathogenic tau species. These variations even exist between individuals that share the same clinical diagnosis (e.g., Alzheimer's disease) and are associated with the disease stage and pathogenicity (2-4).

Characterizing the complex landscape of primary proteinopathies and their secondary effects in neurodegenerative disorders requires brain tissue sampling and histopathological analysis. However, unlike brain tumors, given the potential risks of local injury, invasive brain tissue biopsies are rarely performed for the diagnosis of neurodegenerative disorders (5). Non-invasive detection of brain-derived pathogenic tau molecules and biomarkers of neurodegeneration is imperative to advance the understanding of tauopathies and develop personalized therapeutic strategies.

[0314] Recently, highly sensitive assays have been developed to detect plasma biomarkers of primary proteinopathy and secondary biomarkers that herald downstream processes in neurodegenerative disorders (6,7). Neurofilament light chain protein (NfL) is one of the emerging secondary biomarkers that is released from damaged axons into the extracellular space and subsequently into the bloodstream (8), and has shown promise to predict disease progression (9). However, beyond screening for early detection of neurodegenerative disorders, the clinical utility of these blood-based biomarkers faces crucial challenges. First, the BBB acts as a biased filter with differential permeability to brain-derived biomarkers, limiting detectability of various pathogenic protein species (10,11). Second, fluid biomarkers lack anatomical information on the location of the pathology. Accordingly, variations in disease-related brain-derived biomarkers can be obscured by their release from peripheral tissues or nonspecific release from normal brain tissue. Therefore, conventional approaches are unable to colocalize the source of biomarker release with observed neuroimaging abnormalities, which is key to establish the underlying molecular pathways driving the neurodegenerative changes. Lastly, biomarker levels in biofluids are in a steady state between their production and clearance. Rapid clearance of biomarkers from biofluids may render them undetectable. In addition, the variability caused by inter-individual/intra-individual differences in clearance rates could limit the accuracy of fluid biomarker approaches.

[0315] Sonobiopsy is an emerging technique that has the promise to address these challenges. FUS has the potential to target any area in the whole brain with high precision (on the order of millimeter). Localized FUS combined with microbubbles was first shown to disrupt the BBB in 2001 (12), and has since matured to improve brain drug delivery with its feasibility and safety demonstrated in several clinical trials (13). Sonobiopsy leverages this targeted, reversible, focal opening of the BBB to increase the diagnostic yield of plasma biomarkers originating from anatomically targeted brain location (i.e., spatially selective). By collecting blood samples immediately before and after sonobiopsy, the subject's post-sonobiopsy biomarker levels can be compared against its own pre-sonobiopsy biomarker levels to provide a direct estimate of FUS-induced release from the sonicated brain region. Sonobiopsy can minimize the effect of clearance by collecting the enriched blood sample immediately after biomarker release (i.e., temporally controlled).

[0316] As discussed above, sonobiopsy can be used to facilitate the release of nucleic acid-based biomarkers from brain tumors and brain-derived tissue markers from the healthy brain parenchyma to the bloodstream (14-19). It was hypothesized that the application of sonobiopsy may be extended to neurodegenerative diseases by enhancing the release of tauopathy-associated pathologic tau proteins and markers of neurodegeneration to the blood. In this proof-of-principle study, the capability of sonobiopsy to release phosphorylated tau species (pTau) and NfL into the

bloodstream by opening the BBB was determined.

Results

Sonobiopsy Increased the Normalized Concentrations of Circulating pTau Species in PS19 Mice
[0317] The efficacy of sonobiopsy was first demonstrated to enrich the blood with pathologic pTau species compared with conventional blood-based liquid biopsy. FUS sonication was performed at the right cerebral hemisphere (FIG. 18B), which includes areas that exhibit early pTau deposition in young PS19 mice, i.e. hippocampus, piriform cortex, and entorhinal cortex (20,21). Blood samples were collected via cardiac puncture after either FUS sonication in the sonobiopsy group or no treatment in the control group. The plasma levels of pTau species, phosphorylated at threonine position 181 (pTau-181) and threonine position 231 (pTau-231) (22,23), were measured in FUS-treated PS19 and wild-type mice. In addition, the plasma levels of mouse tau (total mTau) were measured as an internal control to normalize pTau levels in plasma.

[0318] Following FUS sonication, contrast-enhanced T_{sub}.1-weighted MRI was performed to verify and quantify BBB opening (FIG. 18C, FIG. 18D). Except for one wild-type mouse, successful BBB opening was observed in the targeted right hemisphere of all FUS-treated mice (wild-type: 7 out of 8; PS19: 7 out of 7). There was no significant group difference in the volume of FUS-mediated BBB opening and no evidence of off-target BBB opening. The wild-type mouse with no evidence of BBB opening following FUS was excluded from further analysis.

[0319] As expected, the absolute levels of pTau species were higher in the PS19 mice than the wild-type mice (FIG. 19A-FIG. 19C). Though the absolute levels of pTau species were not significantly higher in the FUS-treated PS19 mice (FIG. 19A-FIG. 19C), there was a significant 1.7-fold increase ($p=0.0058$) in the normalized levels of pTau-181 (pTau-181/mTau) from the sonobiopsy-treated PS19 mice (0.58 ± 0.18) compared with the control PS19 mice (0.34 ± 0.10) (FIG. 18E). There was a significant 1.4-fold increase ($p=0.048$) in the normalized levels of pTau-231 (pTau-231/mTau) from the sonobiopsy-treated PS19 mice (0.17 ± 0.06) compared with the control PS19 mice (0.12 ± 0.03) (FIG. 18F). There was no significant increase in normalized pTau-181 or pTau-231 levels between the sonobiopsy and control groups for wild-type mice (FIG. 20A-FIG. 20B). Taken together, these results show that sonobiopsy selectively increased the normalized levels of circulating pTau species in pathologic mice because sonobiopsy had no effect on wild-type mice.

Sonobiopsy Released NfL in the Early Neurodegenerative Stage of PS19 Mice

[0320] The capability of sonobiopsy to release secondary biomarkers of tauopathy was further evaluated. The potential of survival blood collection and the capability to target specific brain regions was also assessed. In the second experiment, it was determined whether sonobiopsy would increase the levels of circulating NfL in the early neurodegenerative stages of PS19 mice (6-month-old) (22). A lower peak negative pressure was applied compared with that used in the first experiment (0.4 MPa vs. 0.68 MPa) and targeted smaller brain areas. Specifically, either the ventral anterolateral cerebral cortex, i.e., olfactory cerebral cortex that includes the piriform cortex and amygdala (FIG. 21A) was targeted, or the hippocampus (FIG. 21D), which both exhibit early neurodegenerative changes in PS19 mice was targeted (22,23). The selection of a smaller target region of the brain enabled examination of the regional specificity of sonobiopsy by evaluating the biomarker release from each localized region. Similar to the first experiment, contrast-enhanced T_{sub}.1-weighted MRI was performed to verify and quantify the volume of BBB disruption. FIG. 21B and FIG. 21E are representative pre-FUS MR images for either the olfactory cerebral cortex or the hippocampus, respectively. FIG. 21C and FIG. 21F are representative post-FUS MR images for each target, respectively. There was no significant difference in the volume of FUS-mediated BBB disruption between mice treated at the olfactory cerebral cortex or the hippocampus.

[0321] To mitigate the effects of variabilities in baseline plasma biomarkers among the PS19 mice, a pre-post study design was adopted to evaluate changes in NfL levels by sampling blood before FUS (sample A, submandibular blood collection), 15 minutes after FUS (sample B, submandibular

blood collection), and 45-60 minutes after FUS (sample C, terminal cardiac puncture). In the control group, a submandibular blood collection (sample A) was followed by a cardiac puncture (sample C). Unlike sample C, NfL measurements of samples A and B in the FUS treated group are directly comparable, as these samples were obtained using the same blood collection method, and plasma NfL levels were measured in the same batch of reagents with the same sample dilutions (TABLE 3). Plasma NfL levels in samples A (pre-sonication) and B (post-sonication) were directly comparable, as the blood samples were obtained using the same blood collection method and plasma NfL levels were measured in the same batch of reagents with the same sample dilutions. TABLE-US-00005 TABLE 3 Plasma sample dilutions for each assay. Assay Experiment Batch dilution mTau 1 — 20x pTau-181 1 — 20x pTau-231 1 — 14x NfL (Samples A/B) 2 Batch 1 60x NfL (Sample C) 2 Batch 2 80x

[0322] There was a 2.3-fold (SD: 0.78) increase in plasma NfL following FUS (sample B vs. sample A: $p=2.2\times 10^{-4}$, $n=8$; FIG. 21G). Next, the ratio of plasma NfL levels in the terminal cardiac sample (sample C) to sample A (submandibular blood; pre-sonication in the FUS-treated group) was calculated. Compared to the control group (non-FUS treated PS19 mice), the sample C to sample A plasma NfL level ratio was 2.6-fold higher in the FUS-treated PS19 mice ($p=0.011$; FUS-treated mice: 3.3 ± 2.2 , $n=9$; control mice: 1.3 ± 0.22 , $n=5$) (FIG. 21H). There was no clear difference in FUS-induced NfL release between the hippocampus and the cortex sonicated groups (FIG. 21G). Altogether, by accounting for the baseline levels of plasma NfL, the findings show that sonication of the hippocampus and cerebral cortex leads to an increase in NfL plasma levels in the PS19 mice even at an early neurodegenerative stage.

Minimal Brain Injury can be Avoided with Localized Sonobiopsy Under Low Peak Negative Pressure

[0323] To detect potential FUS-induced tissue damage and hemorrhage in the targeted brain areas, histological examination and hematoxylin and eosin staining (H&E) staining were performed. Microhemorrhage density was calculated as the percentage of microhemorrhage surface area to the treated hemisphere, using the positive pixel algorithm in QuPath (24). The H&E staining of the 2-month-old PS19 mice (FIG. 22A) revealed that the microhemorrhage density in the treated hemisphere ($1.38\pm 0.92\%$) was significantly greater than that in the contralateral hemisphere ($0.48\pm 0.44\%$; $p=0.04$, $n=8$) (FIG. 22B). For the 6-month-old PS19 mice that were treated with a lower FUS peak negative pressure and a smaller target region, the H&E staining (FIG. 22C) revealed no increase in the microhemorrhage density in the treated hemisphere ($0.52\pm 0.32\%$), compared to the contralateral hemisphere ($0.42\pm 0.29\%$; $p=0.2$, $n=13$) (FIG. 22D). With these optimal sonobiopsy parameters, the microhemorrhage area decreased from 0.051 ± 0.038 mm^{sup.2} to 0.013 ± 0.007 mm^{sup.2} (FIG. 23A-FIG. 23B). In addition, transcranial passive cavitation detection (PCD) was used to monitor microbubble cavitation during focused ultrasound sonication. The inertial cavitation dose, a predictor for FUS-induced tissue damage, was minimal for all FUS-treated groups in experiments 1 and 2 (FIG. 23A-24B). Altogether, these suggest that even minimal brain injury can be avoided by lowering FUS peak negative pressure.

DISCUSSION

[0324] In this study, sonobiopsy significantly enhanced the release of pTau species and secondary markers of neurodegeneration into the bloodstream for noninvasive diagnosis for tauopathies and neurodegenerative diseases. Sonobiopsy leveraged the spatial targeting of FUS to open a relatively small region of the brain, thus allowing for spatially localized biomarker release. Finally, it was shown that sonobiopsy of brain parenchyma is safe, and even minimal brain damage (i.e. microhemorrhage) can be avoided by lowering the peak negative pressure of FUS, while maintaining elevated blood levels of biomarkers beyond baseline measurements.

[0325] Previously, sonobiopsy enabled sensitive detection of circulating biomarkers by enriching the blood samples of mouse and pig glioma models with DNA/RNA-based biomarkers (14,15,17,19). However, the BBB is inherently compromised by tumor growth in these brain tumor

models. A prior study demonstrated the application of sonobiopsy to release normal structural brain proteins in the healthy pig model (16). This study expanded the utility of sonobiopsy by demonstrating the FUS-induced release of both primary pathogenic proteins (pTau species) and downstream biomarkers of neurodegeneration.

[0326] High baseline levels of pTau-181, pTau-231, and NfL were observed. This is expected as these are established biomarkers for tauopathy and neurodegeneration that are readily detectable in plasma with highly sensitive assays. As a result, the apparent yields of sonobiopsy have likely been dampened. It follows that sonobiopsy will have a much higher yield for low abundance peripheral biomarkers, such as DNA/RNA-based biomarkers (14,17). To further complicate diagnostic performance, pTau-181, pTau-231, and NfL had large variabilities across subjects, which likely reflects the heterogeneity observed in neuropathology in PS19 mice (25). Moreover, inter-individual variations in nonspecific release and clearance of biomarkers can result in large differences in baseline levels of brain-derived plasma biomarkers in the steady state. Similarly, in the clinical settings, such variations that are unrelated to the underlying disease process can decrease the sensitivity and specificity of plasma biomarkers in conventional liquid biopsy. In contrast, sonobiopsy can directly address this limitation by allowing blood sample collection before and after sonication to compare the biomarker levels within each subject. In this way, the changes in plasma biomarker levels from their baseline following sonication can be used to directly estimate FUS-induced release of brain-derived biomarkers from the targeted area in the brain.

[0327] The observed microhemorrhage (FIG. 22A) can be avoided by lowering the peak negative pressure (FIG. 22C), while still achieving sonobiopsy-induced biomarker release (FIG. 21G). This suggests that sonobiopsy can be successfully performed without gross brain injury by optimizing FUS parameters. Though the higher peak negative pressure increases the efficacy of sonobiopsy and BBB opening (15,26), the efficacy likely varies depending on the biomarker molecular properties, including the kinetics of biomarker influx into and clearance from the plasma (19). Altogether, there must be a balance between the safety and efficacy of sonobiopsy by optimizing the FUS parameters and plasma sampling times for the biomarker of interest and targeted tissue.

[0328] Though sonobiopsy was effective when targeting the hippocampus or the olfactory cerebral cortex, there was no significant difference in NfL levels between these specific brain regions. This may be explained by early neurodegenerative changes in both targeted areas in PS19 mice (22,23). The spatial resolution of sonobiopsy can be leveraged to colocalize the biomarkers released by sonobiopsy with nonspecific imaging biomarkers of neurodegeneration, such as decreased metabolism (27) or tissue microstructural deficits (28,29). Further, the yield of sonobiopsy can be increased by targeting areas with a higher burden of protein deposits, e.g., tau deposits using tau-specific radiotracers for positron emission tomography (30). This spatially-resolved diagnosis can improve the multimodal diagnosis by combining molecular biomarkers with imaging biomarkers, thus providing an additional benchmark for the accurate diagnosis of the molecular subtypes and regional patterns of neurodegenerative disorders that predict disease progression and outcome.

[0329] This study demonstrated that sonobiopsy could result in a quantitative increase in plasma biomarker levels. This study showed that sonobiopsy can be successfully performed with relatively low FUS acoustic pressures (0.4 MPa) and by targeting a specific brain area.

[0330] In conclusion, this study demonstrated the feasibility and safety of sonobiopsy to enhance the detection of both biomarkers of the primary pathogenic process (e.g., post-translational modifications of the tau protein) and secondary neurodegeneration. As such, sonobiopsy can provide a window into the molecular derangements in the central nervous system milieu that are otherwise hidden or distorted by the BBB to resolve inter-individual variations in brain proteinopathies. This proof-of-principle study was the first to expand the utility of sonobiopsy from brain cancer to neurodegenerative disorders.

Materials and Methods

Animal Preparation

[0331] Two-month-old (first experiment) and 6-month-old (second experiment) male transgenic PS19 (B6;C3-Tg(Prnp-MAPT*P301 S)PS19Vle/J) mice were purchased from the Jackson Laboratory (Stock No: 008169). PS19 mice express P301S human tau (1N4R) under the murine prion promoter (22). Noncarrier wild-type mice were used as controls for the first experiment. The mice were held in a temperature-controlled facility under 12-12 light and dark conditions (light on at 6 AM). They were housed in groups of no more than five. Food and water were provided ad libitum. All animal procedures were reviewed and approved by the Institutional Animal Care and Use Committee at Washington University in St. Louis in accordance with the Guide for the Care and Use of Laboratory Animals and the Animal Welfare Act.

Focused Ultrasound Setup

[0332] The MRI-compatible FUS system used was previously described (17,34). FIG. 18A depicts a schematic of the MRI-guided FUS setup used for sonication of the 2-month-old and 6-month-old mice.

Focused Ultrasound Experimental Setup

[0333] The MRI-compatible FUS transducer (Imasonics, Voray sur l'Ognon, France) was made of a 7-element annular array with a center frequency of 1.5 MHz, an aperture of 25 mm, and a radius of curvature of 20 mm. The annular array design of the FUS transducer allowed it to electronically steer the focus in the axial direction (Z-axis). The transducer was connected to an MRI-compatible piezoelectric motor, allowing the position of the transducer to be mechanically adjusted in the lateral directions (along the X- and Y-axes). The output of the FUS transducer was calibrated using a hydrophone (Onda HGL-0200, Sunnyvale, CA, USA). The axial and lateral full width at half maximums (FWHM) of the FUS transducer were 5.5 mm and 1.2 mm, respectively. Pressure values were derated to account for the 18% mouse skull attenuation (39). The PCD sensor at the center of the FUS transducer had a center frequency of 1.6 MHz and a -6 dB bandwidth of 754 kHz. The signals detected by the PCD sensor were acquired by the PicoScope (5000 series, Pico Technology Ltd, UK). The transducer set (FUS and PCD) was connected to a water balloon filled with deionized and degassed water and coupled to the mouse head with degassed ultrasound gel.

Sonobiopsy Treatment Procedure

[0334] Coronal and axial T₂-weighted fast spin-echo MRI scans were acquired to image the mouse head and locate the geometrical focus of the transducer (TR/TE: 2000/52 ms; slice thickness: 0.5 mm; in-plane resolution: 0.25×0.25 mm²; matrix size: 128×128; averages: 4). The MRI scans were imported to a software program (ThermoGuide, Image Guided Therapy, Pessac, France) to locate the focus of the transducer via 3-point triangulation. For the first experiment, the FUS target covered the hippocampus, piriform cortex, and entorhinal cortex where phosphorylated tau has been observed as young as 1 month of age (20,21). For the second experiment, FUS sonication was performed at either the cerebral cortex or the hippocampus.

[0335] A baseline PCD scan was acquired to ensure sufficient acoustic coupling by the lack of harmonic emission and to validate successful sonication by comparing the change in cavitation activity before and after microbubble injection. Definity microbubbles (Lantheus Medical Imaging, North Billerica, MA) were withdrawn with an additional 18-gauge needle to vent the vial and diluted with preservative-free normal saline to prepare a diluted bolus (microbubble diameter: 1.38±0.13 μm; concentration: 3.87±0.66×10⁹ microbubbles/mL). Microbubbles were injected 30 seconds prior to the start of FUS sonication (frequency: 1.5 MHz, peak negative pressure: 0.68 MPa, pulse repetition frequency: 5 Hz, duty cycle: 3.35%, pulse length: 6.7 ms, treatment duration: 32 sec) to allow the circulation of bubbles to reach a steady-state. To ensure FUS sonication encompasses both the entorhinal cortex and hippocampus, two sonication targets of 4×1 mm were placed with a 1 mm lateral gap between each other. The transducer moved along the anterior-posterior axis to sonicate the medial target first. There was a 5-minute delay before acquiring a second baseline PCD scan and starting FUS sonication of the lateral target to ensure sufficient microbubble clearance (35-37). The PCD signal was processed to calculate the stable and inertial

cavitation doses with a custom MATLAB script (16).

[0336] After sonication, 50 μ L of 1:1 dilution of gadolinium-based MR contrast agent, gadoterate meglumine (Gd-DOTA; Dotarem, Guerbet, Aulnay sous Bois, France) was intravenously injected and followed by a 0.25 mL saline flush. The hyper-enhancement in the T.sub.1-weighted images would indicate BBB disruption. An axial post-FUS coronal T.sub.1-weighted gradient-echo MRI scan (TR/TE: 108/4 ms; slice thickness: 0.5 mm; in-plane resolution: 0.25 $\times$ 0.25 mm.sup.2; matrix size: 128 $\times$ 128; averages: 16) was performed to quantify the FUS-induced changes in BBB permeability. The BBB opening volume was quantified with a custom MATLAB script as previously described (14-19).

BBB Opening Volume Analysis

[0337] First, two regions-of-interest (ROI) were defined in the treated and the contralateral hemispheres. Next, the BBB opening volume was estimated by calculating the sum of voxels, in the treated ROI and across image slices that had a signal intensity greater than 3 $\times$ standard deviations above the mean voxel intensity in the contralateral ROI.

Cavitation Level Analysis

[0338] The stable cavitation level was calculated as the root-mean-squared amplitudes of subharmonic ($f_{\text{sub.0/2}} \pm 0.01$ MHz), ultraharmonic ($3f_{\text{sub.0/2}} \pm 0.01$ MHz), second harmonic ($2f_{\text{sub.0/2}} \pm 0.01$ MHz), and third harmonic ($3f_{\text{sub.0/2}} \pm 0.01$ MHz). The inertial cavitation level was calculated as the root-mean-squared amplitude of the broadband (0.5-4.5 MHz after removing $f_{\text{sub.0/2}} \pm 0.01$ MHz and $nf_{\text{sub.0}} \pm 0.01$ MHz where $n=1, 3/2, 2, 3$) signal.

Plasma Biomarker Detection

[0339] Mouse blood was collected into EDTA tubes and spun at 6,000 g for 5 min at 4° C. The top plasma layer was transferred to a 0.5 ml microcentrifuge tube and stored at -80° C. Frozen plasma samples were thawed at room temperature. All plasma protein measurements were performed in duplicates on a fully automated HD-X Analyzer (Quanterix, Lexington, MA, United States) using ultrahigh sensitive Single Molecule Array (Simoa) kits for mouse tau (Quanterix #102209), pTau-181 (Advantage V2 assay; Quanterix #103714), pTau-231 (Quanterix #102292), and NfL (NF-LIGHT, Quanterix #103186). The plasma sample dilutions for each of these assays are provided in Table E1.

Histological Analysis

[0340] After blood collection, the mice were transcardially perfused with phosphate-buffered saline followed by 4% paraformaldehyde. Mouse brains were collected and fixed to prepare for paraffin sectioning. The whole brain was sectioned horizontally to ensure the treated tissue was captured in each slice. H&E staining was performed to examine red blood cell extravasation and cellular injury. The slices were imaged on the Axio Scan.Z1 Slide Scanner (Zeiss, Oberkochen, Germany). QuPath v0.2.0 (24) was used to detect the presence of FUS-induced hemorrhage. After color deconvolution (hematoxylin vs. eosin), areas of microhemorrhage were detected using the positive-pixel count algorithm. Microhemorrhage density was calculated as the percentage of microhemorrhage surface area over the entire evaluated surface area (%).

Statistical Analysis

[0341] All statistical analyses were performed using R v4.1.2 (<https://www.r-project.org/>). Plots were generated using the ggplot2 package v3.3.5 in R (38). All absolute plasma levels of biomarkers were log-transformed before statistical comparison, while the plasma level ratios were directly compared with no transformation. Two sample t-tests were performed to assess group differences. Paired t-tests were conducted to compare pre/post-sonication levels of NfL and ipsi/contralateral microhemorrhage density. All reported p-values are one-tailed, unless otherwise specified (considered significant at $p < 0.05$).

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Example 5: Enrichment of Tumor-Associated Fragmentomic Features in Plasma with Focused Ultrasound-Enabled Liquid Biopsy in Glioma Patients

[0381] This example shows the utilization of the focused ultrasound for opening the blood brain barrier described herein for diagnosing brain cancer (e.g., glioma).

Objectives

[0382] Blood-based liquid biopsy tests provide a non-invasive means to obtain genomic and epigenomic data. However, circulating tumor DNA (ctDNA) detection in the plasma of glioma patients is difficult due to the presence of the blood-brain barrier (BBB). Focused ultrasound (FUS)-enabled blood-based liquid biopsy (sonobiopsy) locally disrupts the BBB and enriches for brain tumor-derived biomarkers. The enrichment of tumor-associated fragmentomic features with sonobiopsy in patients with glioma was evaluated.

Methods

[0383] Fourteen patients were enrolled prospectively for the clinical study of sonobiopsy with median age of 63 years (range 36-74 years). The sonobiopsy procedure was performed prior to tumor resection, using a nimble FUS device integrated with a clinical neuronavigation system. Blood samples were collected before (5 min pre-FUS) and at different time points post-sonication (5, 10, and 30 min post-FUS). Enzymatic Methyl-seq (EM-Seq) was applied to plasma cell-free DNA (cfDNA) isolated from 14 patients and 10 healthy adults. Tumor tissue obtained from 13 glioma patients was also subjected to EM-Seq and whole genome sequencing. cfDNA fragmentomic features were then quantified, such as the ratio of short fragments (<120 bp) to long fragments (140-250 bp), median fragment length, and coefficient of variation (CV) of the fragment length distribution at the transcription start sites (TSSs) of all protein-coding genes. Tumor fraction

(TFx) levels based on genome-wide copy number alterations in plasma cfDNA was also estimated using ichorCNA.

Results

[0384] Among the 14 glioma patients prospectively enrolled onto this study, 12 were glioblastoma patients and two were low-grade glioma patients. Cell-free DNA sequencing data was analyzed from five glioma patients and 9 healthy adults. Comparing FUS-induced fragmentomic features between pre-FUS and post-FUS samples, the ratio of short to long cfDNA fragments was significantly higher in post-FUS samples ($p < 0.0001$). Similarly, the median cfDNA fragment size was significantly shorter in post-FUS samples compared to pre-FUS ($p < 0.0001$), and post-FUS samples had significantly higher cfDNA fragment size CVs at protein-coding TSSs than pre-FUS ($p < 0.0001$). Furthermore, TFx levels in plasma cfDNA were consistently higher in post-FUS samples compared to pre-FUS.

Conclusions

[0385] cfDNA fragmentation patterns in glioma patients strongly suggest that post-FUS cfDNA is more fragmented than pre-FUS, consistent with the enrichment of brain tumor-derived DNA within the plasma.

Claims

1. A method for diagnosing brain cancer comprising: applying a focused ultrasound to a brain or brain region of a patient, wherein the focused ultrasound is operable to disrupt a blood brain barrier of the patient and release cfDNA across the blood brain barrier of the patient; obtaining a biological sample from the patient, wherein the biological sample includes the cfDNA released across the blood brain barrier; measuring one or more diagnostic fragmentomic features of the cfDNA in the biological sample; comparing the one or more diagnostic fragmentomic features to one or more thresholds; and diagnosing the patient with brain cancer when the one or more diagnostic fragmentomic features exceed the one or more thresholds.
2. The method of claim 1, wherein the one or more diagnostic fragmentomic features include one or more of a ratio of short fragments to long fragments, a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes.
3. The method of claim 1, further comprising obtaining a baseline biological sample prior to applying the focused ultrasound.
4. The method of claim 3, further comprising measuring one or more baseline fragmentomic features in cfDNA in the baseline biological sample.
5. The method of claim 4, wherein the one or more baseline fragmentomic features include one or more of a ratio of short fragments to long fragments, a median fragment length, and a coefficient of variation of a fragment length distribution at transcription sites of all protein-coding genes.
6. The method of claim 5, wherein the one or more thresholds include a percentage increase or decrease between the one or more baseline fragmentomic features and the one or more diagnostic fragmentomic features.
7. The method of claim 1, wherein the one or more thresholds are empirically determined thresholds from a population of healthy patients and/or a population of patients with brain cancer.
8. The method of claim 1, wherein the biological sample is a blood sample.
9. The method of claim 1, wherein the focused ultrasound is applied for a period of time sufficient to disrupt the blood brain barrier.
10. The method of claim 9, wherein the period of time is about 1 minute to about 4 minutes.
11. The method of claim 1, wherein the focused ultrasound is applied at an acoustic pressure sufficient to disrupt the blood brain barrier and release the cfDNA across the blood brain barrier.
12. The method of claim 11, wherein the acoustic pressure is about 0.1 MPa to about 10 MPa.
13. The method of claim 1, further comprising assessing an effectiveness of disruption of the blood

brain barrier or release of cfDNA by measuring MRI contrast before and after applying the focused ultrasound, wherein an increase in the MRI contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

14. The method of claim 1, further comprising assessing an effectiveness of disruption of the blood brain barrier or release of cfDNA by measuring CT contrast before and after applying the focused ultrasound, wherein an increase in the CT contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

15. The method of claim 1, further comprising assessing an effectiveness of disruption of the blood brain barrier or release of cfDNA by measuring ultrasound contrast before and after applying the focused ultrasound, wherein an increase in the ultrasound contrast after applying the focused ultrasound indicates a successful release of cfDNA from the brain or brain region.

16. The method of claim 1, wherein the brain or brain region includes a suspected tumor.

17. The method of claim 1, further comprising administering microbubbles to the patient in an amount sufficient to disrupt the blood brain barrier when applying the focused ultrasound.

18. The method of claim 1, further comprising detecting, via an acoustic sensor, an acoustic signal.

19. The method of claim 18, wherein the acoustic sensor is operable to measure and/or monitor cavitation acoustic emissions.

20. The method of claim 1, wherein the cfDNA includes ctDNA.
